Supplementary material: The Thermal Pure Higgsino at 1.0–1.2 TeV: Relic Mass, Direct-Detection Limits and the LZ 248 keV Event

HEP-Genesis-Agent session record
Project: ramp-research-higgsino-dd

2026-10-22

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `9e56c682`
- Started: 2026-10-16 11:03:59; report generated: 2026-10-22 11:12:25
- Project: ramp-research-higgsino-dd (`/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd`)
- Compute execution site(s) selected in the session: local
- 3 user turn(s), 215 tool call(s) (0 returned errors), 6 figure(s), 13 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
report	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`
coder	`claude-opus-5-5`	`https://mag.genesis.americansciencecloud.org/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`hep-research-aid list_agent_files`	`{}`	`[]`
2	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
3	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
4	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - workspace`
5	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`
6	`local search_arxiv_abstracts`	`{"query": "higgsino dark matter thermal relic 1.1 TeV direct detection", "max_results": 8}`	`[2302.02905v2] Third family quasi-Yukawa unification: Higgsino dark matter, NLSP gluino and all that Abstract: We explore the implications of third family ($t-b-\tau$) quasi-Yukawa unification (QYU) for ...`
7	`local search_arxiv_abstracts`	`{"query": "pure higgsino Sommerfeld enhancement relic abundance mass", "max_results": 5}`	`[1008.2905v2] Thermal abundance of non-relativistic relics with Sommerfeld enhancement Abstract: We propose an analytic treatment for computing the relic abundances of non-relativistic particles whose...`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "higgsino inelastic dark matter mass splitting electroweak loop scattering", "max_results": 6}	[0811.4682v1] Precision Electroweak Measurements and Constraints on the Standard Model Abstract: This note presents constraints on Standard Model parameters using published and preliminary precision...
9	local search_arxiv_ abstracts	{"query": "LZ 2024 WIMP search results spin-independent limits", "max_results": 4}	[1509.02910v2] LUX-ZEPLIN (LZ) Conceptual Design Report Abstract: The design and performance of the LUX-ZEPLIN (LZ) detector is described as of March 2015 in this Conceptual Design Report. LZ is a sec...
10	hep-research-aid search_arxiv	{"query": "pure higgsino dark matter spin-independent cross section two-loop electroweak direct detection", "max_results": 6}	Found 6 papers matching query: 'pure higgsino dark matter spin-independent cross section two-loop electroweak direct detection' (source: arXiv API) Paper 1: arXiv ID: 1901.09751v3 Title: Pseudo Nambu...
11	hep-research-aid search_arxiv	{"query": "LZ dark matter search results 4.2 tonne-year exposure", "max_ results": 4}	Found 4 papers matching query: 'LZ dark matter search results 4.2 tonne-year exposure' (source: arXiv API) Paper 1: arXiv ID: 2410.17036v3 Title: Dark Matter Search Results from 4.2 Tonne-Years of Ex...
12	hep-research-aid search_arxiv	{"query": "XENONnT WIMP search first results nuclear recoil", "max_results": 4}	Found 4 papers matching query: 'XENONnT WIMP search first results nuclear recoil' (source: arXiv API) Paper 1: arXiv ID: 0809.3137v1 Title: The Frontiers of Nuclear Science, A Long Range Plan Authors...

#	Tool	Arguments	Outcome
13	hep-research-aid search_arxiv	{"query": "higgsino dark matter neutrino floor next generation xenon detectors reach", "max_results": 6}	Found 6 papers matching query: 'higgsino dark matter neutrino floor next generation xenon detectors reach' (source: arXiv API) Paper 1: arXiv ID: 1211.7222v1 Title: Direct dark matter detection: the ...
14	local get_arxiv_ metadata	{"arxiv_id": "2609.09830"}	arXiv ID: 2609.09830 Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatterjee, Debottam Das, Syed Adil Pasha, Al...
15	local search_arxiv_ abstracts	{"max_results": 5, "query": "Hisano direct detection electroweak-interacting dark matter wino higgsino gluon contribution"}	[1004.4090v1] A complete calculation for direct detection of Wino dark matter Abstract: In the anomaly-mediated supersymmetry (SUSY) breaking scenario, neutral gaugino of SU(2) _L multiplet, Wino, can ...
16	local search_arxiv_ abstracts	{"max_results": 5, "query": "XENONnT WIMP dark matter search combined science runs 2025"}	[1211.0310v1] Large Synoptic Survey Telescope: Dark Energy Science Collaboration Abstract: This white paper describes the LSST Dark Energy Science Collaboration (DESC), whose goal is the study of dark...
17	local search_arxiv_ abstracts	{"max_results": 4, "query": "Planck 2018 results cosmological parameters"}	[0604069v1] The Scientific Programme of Planck Abstract: For 40 years, the cosmic microwave background (CMB) has been the most important source of information about the geometry and contents of the ...

#	Tool	Arguments	Outcome
18	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
19	local download_arxiv_full_text	{"arxiv_id": "1104.0228", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1104.0228.txt
20	local download_arxiv_full_text	{"arxiv_id": "1411.6930", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1411.6930.txt
21	local search_arxiv_abstracts	{"query": "higgsino spin-independent cross section electroweak loop pure higgsino nucleon", "max_results": 6}	[1812.08019v2] Higgsino Dark Matter in an economical Scherk-Schwarz setup Abstract: We consider a minimal natural supersymmetric model based on an extra dimension with supersymmetry breaking provided ...
22	local download_arxiv_full_text	{"arxiv_id": "2609.09830", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt
23	local search_arxiv_abstracts	{"query": "higgsino relic abundance Sommerfeld bound state thermal mass 1.1 TeV", "max_results": 8}	[1008.2905v2] Thermal abundance of non-relativistic relics with Sommerfeld enhancement Abstract: We propose an analytic treatment for computing the relic abundances of non-relativistic particles whose...

#	Tool	Arguments	Outcome
24	local download_arxiv_full_text	{"arxiv_id": "2410.17036", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2410.17036.txt
25	local download_arxiv_full_text	{"arxiv_id": "2606.31013", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt
26	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1411.6930.txt", "start": 0, "length": 12000}	[chars 0-12000 of 109138] (more available, next start=12000) Title: Heavy neutralino relic abundance with Sommerfeld enhancements - a study of pMSSM scenarios Authors: M. Beneke, C. Hellmann, P. Ruiz...
27	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1411.6930.txt", "start": 30000, "length": 16000}	[chars 30000-46000 of 109138] (more available, next start=46000) iplet minimal dark matter model with the same χ 0 1 mass. In the pure-wino model the mass splitting between the χ 0 1 and χ $\pm$ 1 has to...
28	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1104.0228.txt", "start": 0, "length": 5000}	[chars 0-5000 of 46034] (more available, next start=5000) Title: Direct Detection of Electroweak-Interacting Dark Matter Authors: Junji Hisano, Koji Ishiwata, Natsumi Nagata, Tomohiro Takesako Year: ...

#	Tool	Arguments	Outcome
29	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 14000}	[chars 0-14000 of 71943] (more available, next start=14000) Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatt...
30	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "length": 14000}	[chars 0-14000 of 63686] (more available, next start=14000) Title: Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection Authors: Koichi Hamaguc...
31	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1411.6930.txt", "length": 10000, "start": 46000}	[chars 46000-56000 of 109138] (more available, next start=56000) itive. The enhancement in the χ 0 1χ + 1 channel also saturates as its on-shell production threshold is approached. This should be the...
32	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "start": 14000, "length": 6000}	[chars 14000-20000 of 63686] (more available, next start=20000) andv$\simeq$246 GeVis the Higgs VEV. Additional electroweak oblique parameters introduced in Ref. [48], such as?U,V, andX, are not generated a...
33	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "start": 30000, "length": 14000}	[chars 30000-44000 of 63686] (more available, next start=44000) g2 tan2θW 120π2 M 2 W µ2 .(39) These expressions imply the simple relation Y$\simeq$tan 2θWW ,(40) which explains the nearly linear correlatio...

#	Tool	Arguments	Outcome
34	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1104.0228.txt", "length": 9000, "start": 22000}	[chars 22000-31000 of 46034] (more available, next start=31000) (n2 - (4Y 2 + 1))/ 8 to their result of the Wino DM case. (as the ?rst term in Eq. (22)). Even for the Z-boson exchange contribution,...
35	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "length": 10000, "start": 20000}	[chars 20000-30000 of 63686] (more available, next start=30000) .48 1.55) ×10-5 ,(26) and ρ= ? ??????? 1 0.856 0.315 0.672 0.856 1 0.154 0.352 0.315 0.154 1 0.221 0.672 0.352 0.221 1 ? ??????? ,(27)...
36	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 14000, "start": 14000}	[chars 14000-28000 of 71943] (more available, next start=28000) 4) whereAandZrepresent the mass number and the atomic number of the target nucleus, respec-tively,µ p stands for the reduced mass of t...
37	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1104.0228.txt", "length": 5000, "start": 31000}	[chars 31000-36000 of 46034] (more available, next start=36000) [(n2 - (4 Y2 + 1))/8] ×10-9 Higgs mass [GeV] gluon(W) -1 -0.5 0 0.5 1 115 200 300 400 500 fp/mp in units of Y2 ×10-9 Higgs ma...
38	local search_arxiv_abstracts	{"max_results": 8, "query": "electroweak doublet dark matter thermal mass higgsino 1.1 TeV relic abundance bound state formation"}	[1407.7874v2] Bound-state formation for thermal relic dark matter and unitarity Abstract: We show that the relic abundance of thermal dark matter annihilating via a long-range interaction, is signific...

#	Tool	Arguments	Outcome
39	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 16000, "start": 28000}	[chars 28000-44000 of 71943] (more available, next start=44000) inally, we investigate whether the inelastic rate can exceed the elastic rate to be the domi- nant DD channel in some parameter space r...
40	hep-research-aid get_paper_figures	{"paper_id": "1411.6930", "figures": "7", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/figures_1411.6930/fig07.png"], "message": "Extracted 1 of 14 figures from... [Fig. 6]
41	local search_arxiv_abstracts	{"max_results": 4, "query": "Hill Solon Standard Model anatomy WIMP dark matter direct detection heavy higgsino cross section"}	[1912.07795v2] Direct detection rate of heavy Higgsino-like and Wino-like dark matter Abstract: A large class of viable dark matter models contain a WIMP candidate that is a component of a new electro...
42	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 12000, "start": 44000}	[chars 44000-56000 of 71943] (more available, next start=56000) n. Additionally, the inelastic event rate is Boltzmann suppressed due to the requirement of a minimum -- Page 16 -- 16 TABLE VII. Nu...
43	local search_arxiv_abstracts	{"max_results": 6, "query": "Closing the window on WIMP dark matter thermal mass electroweak multiplets bound states"}	[0509008v3] Precision Electroweak Measurements on the Z Resonance Abstract: We report on the final electroweak measurements performed with data taken at the Z resonance by the experiments operating ...

#	Tool	Arguments	Outcome
44	local get_arxiv_ metadata	{"arxiv_id": "2607.20588"}	arXiv ID: 2607.20588 Title: Unveiling the Vanishing Higgsino-Nucleon Scattering in the MSSM at Next-to-Leading Order Authors: Subhadip Bisal, Arindam Chatterjee, Debottam Das, Syed Adil Pasha, Rahul P...
45	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 2410.17036.txt", "length": 10000, "start": 0}	[chars 0-10000 of 63958] (more available, next start=10000) Title: Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment Authors: J. Aalbers, D. S. Akerib, A. ...
46	local download_arxiv_ full_text	{"arxiv_id": "hep-ph/ 0610249", "output_dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ hep-ph_0610249.txt
47	local download_arxiv_ full_text	{"arxiv_id": "1912.07795", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 1912.07795.txt
48	board post_finding	{"claim": "Kinematic threshold for inelastic up-scattering on xenon (my arithmetic): m_A = 131 x 0.9315 GeV = 122.0 GeV; mu_A = 108.7 GeV (m=1.0 TeV) and 109.8 GeV (m=1.1 TeV); v_max = 544+232 = 776 km/s, beta^2 = (776/2...	Recorded finding f-001.

#	Tool	Arguments	Outcome
49	board post_finding	{"claim": "2609.09830 does not quote an LZ-excluded delta_m0 range. Its five benchmarks (|mu| = 0.61-1.40 TeV; |M1|, M2 about 5.8-42 TeV; tan beta 10.9-26.5) have one-loop on-shell splittings delta = 8.9-36.4 keV. Each p...	Recorded finding f-002.
50	board post_finding	{"claim": "2606.31013: the SI coupling comes from Higgs exchange, f_q ~ -(g^2/8M_ h^2)[1/(M2-|mu|) + tan^ 2 theta_W/(M1-|mu|)](1 + sgn(mu) sin 2beta), so it is suppressed at large gaugino mass and at small tan beta with mu...	Recorded finding f-003.
51	local download_arxiv_ full_text	{"arxiv_id": "1807.06209", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 1807.06209.txt
52	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 2410.17036.txt", "length": 5000, "start": 10000}	[chars 10000-15000 of 63958] (more available, next start=15000) oils induced by weakly interacting massive particle (WIMP) dark matter using the LUX-ZEPLIN (LZ) two-phase xenon time projection chambe...
53	local cite	{"arxiv_id": "2609.09830", "note": "[\"mass splitting$\sim$O(10-100) MeV or even smaller, for very large gaugino mass parameters\", \"In scenarios where ?χ0\", \"All benchmark points are consistent with the bounds from the LZ...	Cited: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter (\cite{arxiv260909830})

#	Tool	Arguments	Outcome
54	local cite	{"arxiv_id": "2606.31013", "note": "[\"dark matter direct detection experiments are particularly sensitive to scenarios with larger charged-neutral mass splittings induced by higgsino-gaugino mixing\", \"This expression ...	Cited: Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection (\cite{arxiv260631013})
55	board post_question	{"text": "The tree-level formula delta_m0 ≈ m_Z^2 s_W^2/M1 + m_Z^2 c_W^2/M2 has not been sourced yet. Can a quotable reference be found, for example Ref. [28] of 2609.09830, Eq. (9)? Taken at face value with same-sign M2...	Recorded question q-05.
56	board post_finding	{"claim": "Hisano et al. (1104.0228) find the leading-order electroweak-loop SI cross section of pure EW-multiplet DM (including the higgsino, n=2, Y=1/2, evaluated at M=1 TeV) to be O(1e-46 to 1e-48) cm^2, suppressed by...	Recorded finding f-004.
57	board post_finding	{"claim": "LZ 2024 (2410.17036, 4.2 tonne-years): the strongest SI exclusion is 2.2e-48 cm^2 (90% CL) and the best median sensitivity is 5.1e-48 cm^2, both at 40 GeV/c^2. The abstract quotes no value at 1 TeV.", "evidenc...	Recorded finding f-005.
58	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt", "length": 4000, "start": 0}	[chars 0-4000 of 37078] (more available, next start=4000) Title: Direct detection rate of heavy Higgsino-like and Wino-like dark matter Authors: Qing Chen, Richard J. Hill Year: 2019 arXiv ID: 1912.0...

#	Tool	Arguments	Outcome
59	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt", "length": 6000, "start": 24000}	[chars 24000-30000 of 37078] (more available, next start=30000) tially have a larger than expected impact. We investigate this possibility here, using the model presented in Refs. [28, 29] which incl...
60	board post_finding	{"claim": "No relic-density or inelastic-DM compute tool and no input data tables are available: list_input_files returned an empty file list, and the catalog contains only statistics/histogram, test-job and literature t...	Recorded finding f-006.
61	board post_finding	{"claim": "Beneke, Hellmann and Ruiz-Femenia (arXiv:1411.6930, Sec. 3) report Omega h^2 = 0.127 (perturbative) and 0.126 (Sommerfeld, with off-diagonal rates) for a pure-higgsino model compared against the higgsino-like ...	Recorded finding f-007.
62	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/hep-ph_0610249.txt", "length": 6000, "start": 0}	[chars 0-6000 of 21259] (more available, next start=6000) Title: Non-perturbative Effect on Thermal Relic Abundance of Dark Matter Authors: Junji Hisano, Shigeki Matsumoto, Minoru Nagai, Osamu Saito,...
63	board post_finding	{"claim": "Chen and Hill (1912.07795) include the O(mW/M) power corrections, perturbative QCD to NNNLO and ChEFT nuclear effects. They find that the pure-higgsino SI cross section has an upper limit below 1e-48 cm^2 in t...	Recorded finding f-008.

#	Tool	Arguments	Outcome
64	local cite	{"arxiv_id": "1912.07795", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/references.bib", "note": "[\"The upper limit for the pure doublet is less than 10-48 c...	Cited: Direct detection rate of heavy Higgsino-like and Wino-like dark matter (\cite{arxiv191207795})
65	local cite	{"arxiv_id": "2410.17036", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/references.bib", "note": "[\"The strongest SI exclusion set is 2 .2 × 10-48 cm2 at the...	Cited: Dark Matter Search Results from 4.2 Tonne-Years of Exposure of the LUX-ZEPLIN (LZ) Experiment (\cite{arxiv241017036})
66	local cite	{"arxiv_id": "1411.6930", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/references.bib", "note": "[\"the relic abundances in the pure-higgsino model, obtained ...	Cited: Heavy neutralino relic abundance with Sommerfeld enhancements - a study of pMSSM scenarios (\cite{arxiv14116930})
67	board post_question	{"text": "Still open from d-02: the XENONnT 2303.14729 SI limit at 1 TeV, the actual LZ 2024 limit-curve value at 1 TeV (read from its figure rather than linearly rescaled from 40 GeV), and a numeric xenon neutrino-fog v...	Recorded question q-06.

#	Tool	Arguments	Outcome
68	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/hep-ph_0610249.txt", "length": 6000, "start": 13000}	[chars 13000-19000 of 21259] (more available, next start=19000) the non- perturbative e?ect when m is larger than $\sim$ 1 TeV. A signi?cant enhancement is shown at m $\sim$ 2.4 GeV. This originates in the bou...
69	board post_finding	{"claim": "Hisano et al. (hep-ph/0610249) expect the Sommerfeld effect on the higgsino-like relic abundance to be roughly 10 percent, compared with a 50 percent reduction for the wino. 1411.6930 later found the net pure-...	Recorded finding f-009.
70	local get_arxiv_metadata	{"arxiv_id": "2606.31013"}	arXiv ID: 2606.31013 Title: Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection Authors: Koichi Hamaguchi, Natsumi Nagata, Genta Osaki Year: 20...
71	local get_arxiv_metadata	{"arxiv_id": "2609.09830"}	arXiv ID: 2609.09830 Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatterjee, Debottam Das, Syed Adil Pasha, Al...
72	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt", "start": 0, "length": 4000}	[chars 0-4000 of 442128] (more available, next start=4000) Title: Planck 2018 results. VI. Cosmological parameters Authors: Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Bac...

#	Tool	Arguments	Outcome
73	hep-research-aid get_paper_figures	{"paper_id": "2410.17036", "max_figures": 6, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/figures_2410.17036/fig01.png", "/Users/nramachandra/hep-genesis-testing/p... [Fig. 10, 11]
74	local search_arxiv_abstracts	{"query": "higgsino relic abundance bound-state formation Sommerfeld thermal mass TeV", "max_results": 8}	[1611.08133v2] Effects of QCD bound states on dark matter relic abundance Abstract: We study scenarios where there exists an exotic massive particle charged under QCD in the early Universe. We calcula...
75	hep-research-aid get_paper_figures	{"paper_id": "2303.14729", "max_figures": 6, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437"}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/figures_2303.14729/fig01.png", "/Users/nramachandra/hep-genesis-testing/p... [Fig. 12, 13, 14]
76	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt", "start": 0, "length": 3000}	[chars 0-3000 of 37078] (more available, next start=3000) Title: Direct detection rate of heavy Higgsino-like and Wino-like dark matter Authors: Qing Chen, Richard J. Hill Year: 2019 arXiv ID: 1912.0...

#	Tool	Arguments	Outcome
77	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "start": 8000, "length": 14000}	[chars 8000-22000 of 63686] (more available, next start=22000) ty between elec- troweak precision measurements and dark matter direct detection experiments in probing light higgsinos. Finally, Sec. 6...
78	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "start": 5000, "length": 12000}	[chars 5000-17000 of 71943] (more available, next start=17000) 7, 38] presented the SUSY QCD corrections for the DD of ?χ0 1 DM.2 The DM- nucleon scattering cross-section at one-loop level for a gene...
79	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 10000, "start": 17000}	[chars 17000-27000 of 71943] (more available, next start=27000) e parton-level Lagrangian as, f inel N mN = X u,d, s λ? qfN q - 8π 9αs λ? gfN g (10) Note that, for simplicity, we consider scalar contr...
80	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "length": 12000, "start": 24000}	[chars 24000-36000 of 63686] (more available, next start=36000) elements and a detailed discussion of the associated theoretical uncertainties in dark matter-nucleon scattering cross sections, see R...
81	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt", "start": 22000, "length": 7000}	[chars 22000-29000 of 37078] (more available, next start=29000) 2m2 c, and 1 .0 GeV$\leq$ µ0 $\leq$ 1.4 GeV, as in Ref. [18, 27] . Uncertainties from neglected 1/M2 and higher order power corrections are estima...

#	Tool	Arguments	Outcome
82	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt", "start": 5500, "length": 4500}	[chars 5500-10000 of 442128] (more available, next start=10000) D. Spencer82, R. Sunyaev72, 91, A.-S. Suur-Uski24, 40, J. A. Tauber36, D. Tavagnacco43,33, M. Tenti46, L. To?olatti17, 41, M. Tomasi32,45...
83	local search_arxiv_abstracts	{"max_results": 8, "query": "last complex WIMPs standing electroweak multiplet thermal mass bound states"}	[0811.4682v1] Precision Electroweak Measurements and Constraints on the Standard Model Abstract: This note presents constraints on Standard Model parameters using published and preliminary precision...
84	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt", "length": 2200, "start": 21900}	[chars 21900-24100 of 63686] (more available, next start=24100) dark matter and nucleon masses, respectively, andfN is the effective SI coupling between dark matter and the nucleon. For higgsino dar...
85	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt", "length": 14000, "start": 27000}	[chars 27000-41000 of 71943] (more available, next start=41000) lision will be similar to the elastic one up to a factor proportional to the square of ?χ0 1 ?χ0 2hj couplings. • Consequently, we compu...
86	local search_arxiv_abstracts	{"max_results": 6, "query": "XLZD design book sensitivity spin-independent neutrino fog TeV"}	[0211001v1] Very Long Baseline Neutrino Oscillation Experiment for Precise Determination of Oscillation Parameters and Search for nu_mu -> nu_e Appearance and CP Violation Abstract: The possibility ...

#	Tool	Arguments	Outcome
87	local get_arxiv_ metadata	{"arxiv_id": "2107.09688"}	arXiv ID: 2107.09688 Title: Closing the window on WIMP Dark Matter Authors: Salvatore Bottaro, Dario Buttazzo, Marco Costa, Roberto Franceschini, Paolo Panci, Diego Redigolo, Ludovico Vittorio Year: 2...
88	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 2410.17036.txt", "length": 8000, "start": 20000}	[chars 20000-28000 of 63958] (more available, next start=28000) the tuning of the nest 2.4.0 [18] model for detector response, in which the scin- tillation photon and ionization electron gains, g1 an...
89	local get_arxiv_ metadata	{"arxiv_id": "2205.04486"}	arXiv ID: 2205.04486 Title: The last Complex WIMPs standing Authors: Salvatore Bottaro, Dario Buttazzo, Marco Costa, Roberto Franceschini, Paolo Panci, Diego Redigolo, Ludovico Vittorio Year: 2022 Abs...
90	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437/ 2609.09830.txt", "length": 9000, "start": 41000}	[chars 41000-50000 of 71943] (more available, next start=50000) s-section more than the corresponding elastic cross- section through constructive interference. As a result, inelastic scattering sligh...
91	board post_finding	{"claim": "Chen and Hill (1912.07795), verbatim from the Summary: \"The upper limit for the pure doublet is less than 10-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than ...	Recorded finding f-010.

#	Tool	Arguments	Outcome
92	board post_finding	{"claim": "Correction to f-008: an upper bound below 1e-48 cm^2 compared with the LZ minimum of 2.2e-48 cm^ 2 (at 40 GeV) is a factor of more than 2.2 (more than 0.34 decades), not 2 orders of magnitude. The comparison th...	Recorded finding f-011.
93	board post_dead_end	{"what": "Reading the LZ 2024 (2410.17036) and XENONnT (2303.14729) SI limits at 1 TeV with get_ paper_figures", "why": "The HTML extraction returned none of the limit plots. For LZ it returned only Fig. 1 (fiducial-volum...	Recorded dead end x-01.
94	board post_question	{"text": "The XLZD Design Book (arXiv:2410.17137) has been identified but not read: what are its projected SI sensitivity and its xenon neutrino-fog value at about 1 TeV, and how do they compare with the pure-higgsino bo...	Recorded question q-07.
95	board post_finding	{"claim": "Planck 2018 (arXiv:1807.06209, abstract, page 1) states verbatim: \ "A combined analysis gives dark matter density Ωch2 = 0.120± 0.001\" (68% confidence region), quoted for base-LambdaCDM from polarization, tem...	Recorded finding f-012.
96	board post_finding	{"claim": "Bottaro et al., \"The last Complex WIMPs standing\" (arXiv: 2205.04486, 2022), is the independent recent calculation of thermal masses for complex electroweak n-plets, which includes the Y=1/2 doublet (the higg...	Recorded finding f-013.

#	Tool	Arguments	Outcome
97	board post_question	{"text": "What thermal mass, with its theory uncertainty, does arXiv: 2205.04486 (Bottaro et al. 2022, Sommerfeld plus bound-state formation) give for the Y=1/2 doublet in its thermal-mass table, and how far is it from th...	Recorded question q-08.
98	board post_dead_end	{"what": "Searching abstracts for a recent higgsino thermal mass with the query 'last complex WIMPs standing electroweak multiplet thermal mass bound states'.", "why": "The search returned only unrelated LEP electroweak-...	Recorded dead end x-02.
99	board post_finding	{"claim": "Metadata confirms both IDs: arXiv: 2606.31013 is 'Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection' by Hamaguchi, Nagata and Osaki (2026), and arXiv:2...	Recorded finding f-014.
100	board post_finding	{"claim": "The full text of 2606.31013 supports the f-003 numbers. Eq. 13 reads Delta m+|rad ~ (1/2) alpha2 MZ sin^2 thetaW (1 - 3MZ/(2 pi M_chi+)), with the statement 'For M_chi+ > > MZ, this mass splitting asymptoticall...	Recorded finding f-015.
101	board post_finding	{"claim": "The full text of 2609.09830 supports the f-002 numbers. Table III gives one-loop on-shell delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5, with mu = -739.63, -606.64, -814.26, 1282.11 and 1400.0 Ge...	Recorded finding f-016.

#	Tool	Arguments	Outcome
102	board post_finding	{"claim": "2609.09830 does not write out the tree-level delta_m0 formula. It says only: 'as can be seen, for example, from Eq. (9) of Ref. [28], the corrections proportional to M_Z^2/M1 and M_Z^2/M2 depend on the relativ...	Recorded finding f-017.
103	board post_question	{"text": "What arXiv paper is Ref. [28] of 2609.09830, and does its Eq. (9) give the tree-level neutral higgsino splitting with explicit sign dependence on mu, M1 and M2? This is needed to re-evaluate the M1 required for...	Recorded question q-09.
104	hep-research-aid list_agent_files	{}	[]
105	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
106	hep-research-aid list_datasets	{}	{'count': 0, 'datasets': []}
107	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...
108	hep-research-aid get_dispatch	{}	Dispatch: local execution.
109	local search_arxiv_abstracts	{"query": "higgsino thermal relic 1.1 TeV dark matter direct detection", "max_results": 8}	[2302.02905v2] Third family quasi-Yukawa unification: Higgsino dark matter, NLSP gluino and all that Abstract: We explore the implications of third family ($t-b-\tau$) quasi-Yukawa unification (QYU) for ...

#	Tool	Arguments	Outcome
110	local search_arxiv_ abstracts	{"query": "pure higgsino dark matter mass splitting inelastic scattering LZ XENONnT", "max_results": 8}	[1509.02910v2] LUX-ZEPLIN (LZ) Conceptual Design Report Abstract: The design and performance of the LUX-ZEPLIN (LZ) detector is described as of March 2015 in this Conceptual Design Report. LZ is a sec...
111	local search_arxiv_ abstracts	{"query": "Sommerfeld enhancement higgsino relic abundance bound state formation", "max_results": 6}	[0910.5221v2] Relic density and CMB constraints on dark matter annihilation with Sommerfeld enhancement Abstract: We calculate how the relic density of dark matter particles is altered when their an...
112	local get_arxiv_ metadata	{"arxiv_id": "2609.01590"}	arXiv ID: 2609.01590 Title: TeV Higgsino Dark Matter from LZ Nuclear Recoil to Fermi-LAT Gamma Rays Authors: Lei Wu, Yang Zhang, Bin Zhu Year: 2026 Abstract: The excess of high-energy nuclear recoils,...
113	local get_arxiv_ metadata	{"arxiv_id": "2609.07807"}	arXiv ID: 2609.07807 Title: Not so good νs for Higgsino dark matter as LZ excess: stringent limits from Super-Kamiokande and IceCube Authors: Debajit Bose, Akash Kumar Saha, Deep Jyoti Das, Rinchen ...
114	local get_arxiv_ metadata	{"arxiv_id": "2609.09830"}	arXiv ID: 2609.09830 Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatterjee, Debottam Das, Syed Adil Pasha, Al...

#	Tool	Arguments	Outcome
115	local search_arxiv_ abstracts	{"max_results": 6, "query": "LZ WIMP search results exposure spin-independent limit 2024"}	[1509.02910v2] LUX-ZEPLIN (LZ) Conceptual Design Report Abstract: The design and performance of the LUX-ZEPLIN (LZ) detector is described as of March 2015 in this Conceptual Design Report. LZ is a sec...
116	hep-research-aid search_arxiv	{"query": "LZ dark matter search results 4.2 tonne-year exposure", "max_ results": 5}	Found 5 papers matching query: 'LZ dark matter search results 4.2 tonne-year exposure' (source: arXiv API) Paper 1: arXiv ID: 2410.17036v3 Title: Dark Matter Search Results from 4.2 Tonne-Years of Ex...
117	local search_arxiv_ abstracts	{"query": "XENONnT WIMP dark matter search nuclear recoil results", "max_ results": 5}	[0504031v1] Limits on WIMP cross-sections from the NAIAD experiment at the Boulby Underground Laboratory Abstract: The NAIAD experiment (NaI Advanced Detector) for WIMP dark matter searches at the B...
118	local search_arxiv_ abstracts	{"query": "electroweak multiplet dark matter thermal mass higgsino 1.1 TeV wino 2.8 TeV", "max_ results": 6}	[1901.02987v2] Discovery reach for wino and higgsino dark matter with a disappearing track signature at a 100 TeV pp collider Abstract: Within the theory of supersymmetry, the lightest neutralino is...
119	local search_arxiv_ abstracts	{"query": "LZ high energy nuclear recoil event inelastic dark matter higgsino mass splitting", "max_results": 6}	[0510346v1] The Dark Energy Survey Abstract: We describe the Dark Energy Survey (DES), a proposed optical-near infrared survey of 5000 sq. deg of the South Galactic Cap to ~24th magnitude in SDSS gr...

#	Tool	Arguments	Outcome
120	local search_arxiv_ abstracts	{"max_results": 5, "query": "LZ collaboration search inelastic dark matter high recoil energy nuclear recoil event 248 keV"}	[1211.0310v1] Large Synoptic Survey Telescope: Dark Energy Science Collaboration Abstract: This white paper describes the LSST Dark Energy Science Collaboration (DESC), whose goal is the study of dark...
121	engine describe_tools	{}	{ "server": "inelastic-dm-server", "description": "Bridges particle physics (WIMP mass, mass splitting, cross section) with cosmological observables (relic density, recoil spectra, indirect detect...
122	local download_arxiv_ full_text	{"arxiv_id": "2609.01590", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244/ 2609.01590.txt
123	serve-idm.sh compute_relic_ density	{"m_chi_GeV": 900}	{ "parameters": { "m_chi_GeV": 900.0, "is_higgsino": true, "sigma_v_cm3_s": 3.7345679012345686e-26 }, "relic_density": { "omega_h2": 0.0731, "planck_value": 0.12, "planck...
124	serve-idm.sh compute_relic_ density	{"m_chi_GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "is_higgsino": true, "sigma_v_cm3_s": 3.0250000000000006e-26 }, "relic_density": { "omega_h2": 0.0898, "planck_value": 0.12, "planc...

#	Tool	Arguments	Outcome
125	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1100}	{ "parameters": { "m_chi_GeV": 1100.0, "is_higgsino": true, "sigma_v_cm3_s": 2.5e-26 } , "relic_density": { "omega_h2": 0.1083, "planck_value": 0.12, "planck_error": 0.001...
126	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1200}	{ "parameters": { "m_chi_GeV": 1200.0, "is_higgsino": true, "sigma_v_cm3_s": 2.1006944444444443e-26 }, "relic_density": { "omega_h2": 0.1285, "planck_value": 0.12, "planc...
127	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1300}	{ "parameters": { "m_chi_GeV": 1300.0, "is_higgsino": true, "sigma_v_cm3_s": 1.789940828402367e-26 }, "relic_density": { "omega_h2": 0.1503, "planck_value": 0.12, "planck...
128	local download_arxiv_full_text	{"arxiv_id": "2609.07807", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.07807.txt
129	engine describe_tools	{}	{ "server": "inelastic-dm-server", "description": "Bridges particle physics (WIMP mass, mass splitting, cross section) with cosmological observables (relic density, recoil spectra, indirect detect...

#	Tool	Arguments	Outcome
130	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 300}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 300.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 270.0, "v_min_at_p...
131	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 400}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 400.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 360.1, "v_min_at_p...
132	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 480}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 480.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 432.1, "v_min_at_p...
133	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 520}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 520.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 468.1, "v_min_at_p...
134	serve-idm.sh predict_recoil	{"m_chi_GeV": 1100, "delta_keV": 560}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 560.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 504.1, "v_min_at_p...

#	Tool	Arguments	Outcome
135	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.01590.txt", "length": 20000}	[chars 0-20000 of 58419] (more available, next start=20000) Title: TeV Higgsino Dark Matter from LZ Nuclear Recoil to Fermi-LAT Gamma Rays Authors: Lei Wu, Yang Zhang, Bin Zhu Year: 2026 arXiv ID: 26...
136	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.07807.txt", "length": 20000}	[chars 0-20000 of 33556] (more available, next start=20000) Title: Not so good νs for Higgsino dark matter as LZ excess: stringent limits from Super-Kamiokande and IceCube Authors: Debajit Bose, Ak...
137	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1150}	{ "parameters": { "m_chi_GeV": 1150.0, "is_higgsino": true, "sigma_v_cm3_s": 2.287334593572779e-26 }, "relic_density": { "omega_h2": 0.1181, "planck_value": 0.12, "planck...
138	serve-idm.sh compute_relic_density	{"m_chi_GeV": 1160}	{ "parameters": { "m_chi_GeV": 1160.0, "is_higgsino": true, "sigma_v_cm3_s": 2.2480677764565993e-26 }, "relic_density": { "omega_h2": 0.1202, "planck_value": 0.12, "planc...
139	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.01590.txt", "start": 20000, "length": 12000}	[chars 20000-32000 of 58419] (more available, next start=32000) lines [17]. We compute the complete NLO higgsinoγ+X -- Page 5 -- 5 950 1000 1050 1100 1150 1200 mχ0 1 [GeV] 0.8 1.0 1.2 1.4 1.6 Hig...

#	Tool	Arguments	Outcome
140	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.07807.txt", "start": 20000, "length": 9000}	[chars 20000-29000 of 33556] (more available, next start=29000) direct detectors such as LZ, XENONnT, and PandaX-4T. We re-state that our results do not completely rule out the thermal Higgsino inter...
141	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1160, "delta_keV": 500}	{ "parameters": { "m_chi_GeV": 1160.0, "delta_keV": 500.0, "sigma_n_cm2": 7.430586727243295e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1202, "tension_si...
142	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1160, "delta_keV": 400}	{ "parameters": { "m_chi_GeV": 1160.0, "delta_keV": 400.0, "sigma_n_cm2": 7.430586727243295e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1202, "tension_si...
143	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1160, "delta_keV": 600}	{ "parameters": { "m_chi_GeV": 1160.0, "delta_keV": 600.0, "sigma_n_cm2": 7.430586727243295e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1202, "tension_si...
144	hep-research-aid get_dispatch	{}	Dispatch: local execution.
145	serve-idm.sh scan_mass_splitting	{"m_chi_min": 900, "m_chi_max": 1300, "delta_min": 300, "delta_max": 650, "n_points": 15}	{ "scan_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz", "grid": "15 x 15 = 225 points", "ranges": { "m_chi_GeV": [900.0, 1300.0], ...

#	Tool	Arguments	Outcome
146	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.01590.txt", "start": 12000, "length": 8000}	[chars 12000-20000 of 58419] (more available, next start=20000) - fore directly testable by IceCube searches for solar neu- trinos [15] . The captured higgsinos initially lose orbital energy through i...
147	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.07807.txt", "start": 12000, "length": 8000}	[chars 12000-20000 of 33556] (more available, next start=20000) i d is the measured ν + $\bar{}\nu$ flux with uncertaintyσi d, Φi atm is the atmospheric neutrino model flux (with appropriate angular cuts) , an...
148	serve-idm.sh plot_recoil_spectrum	{"m_chi_GeV": 1100, "delta_keV": 500}	{ "plot_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/recoil_spectrum.png", "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 500.0 } }
149	serve-idm.sh predict_annual_modulation	{"m_chi_GeV": 1100, "delta_keV": 500, "E_R_keV": 248}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 500.0, "E_R_keV": 248.0 }, "modulation": { "peak_day": 1, "peak_date": "January 01", "trough_day": 1, "trough_date...
150	serve-idm.sh check_all_constraints	{"m_chi_GeV": 1100, "delta_keV": 500}	{ "parameters": { "m_chi_GeV": 1100.0, "delta_keV": 500.0, "sigma_n_cm2": 7.429930732553727e-39, "is_higgsino": true } , "relic_density": { "omega_h2": 0.1083, "tension_si...

#	Tool	Arguments	Outcome
151	local cite	{"arxiv_id": "2609.01590", "note": "[\"including the high-velocity tail generated by the gravitational influence of the Large Magellanic Cloud yields\", \"The profiled Maxwellian likelihood peaks atm χ = 1.09 TeV andδ m ...	QUOTE FAILED: "The upper reach inδ m is controlled primarily by the kinematic endpointv esc +vE" not found in source
152	local cite	{"arxiv_id": "2609.07807", "note": "[\"excluding inter-state mass splittings ? 557 keV. This disfavors Higgsino-like interpretations of the LZ event for standard halo velocities.\", \"with the circular speed u0 = 220 km/ ...	Cited: Not so good $$\nu$$s for Higgsino dark matter as LZ excess: stringent limits from Super-Kamiokande and IceCube (\ cite{arxiv260907807})
153	local get_arxiv_ metadata	{"arxiv_id": "2610.03692"}	arXiv ID: 2610.03692 Title: Higgsino dark matter compatible with the LUX-ZEPLIN high-energy nuclear-recoil event and IceCube constraints Authors: Katherine Freese, Dionysios P. Theodosopoulos Year: 20...
154	local get_arxiv_ metadata	{"arxiv_id": "2609.09830"}	arXiv ID: 2609.09830 Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatterjee, Debottam Das, Syed Adil Pasha, Al...
155	local get_arxiv_ metadata	{"arxiv_id": "2609.06890"}	arXiv ID: 2609.06890 Title: High-Energy Nuclear Recoils from Boosted Dark Matter for the LZ 248-keV Event: Beyond the Halo-Dependent High-Velocity Tail Authors: Haider Alhazmi, Doojin Kim, Kyoungchul ...

#	Tool	Arguments	Outcome
156	board post_finding	{"claim": "At m_chi=1100 GeV (higgsino, sigma_n= 7.43e-39 cm^2, hard halo cutoff v_esc+v_E=776 km/s) , predict_recoil gives v_ min(248 keV)=701/823/921/ 969/1018 km/s and E_R,peak= 270/360/432/468/504 keV for delta=300/400/48...	Recorded finding f-018.
157	board post_finding	{"claim": "Local scan_ mass_splitting (dispatch reported 'local execution') over m_chi=900-1300 GeV, delta=300-650 keV (15x15) finds 180/225 points pass all constraints and only 12 predict 0.3-5 LZ events; check_all_const...	Recorded finding f-019.
158	serve-idm.sh plot_constraint_ summary	{"scan_file": "/ Users/nramachandra/ github-integration/ inelastic-dm-server/output/ scan_idm.npz"}	{ "plot_file": "/ Users/nramachandra/ github-integration/ inelastic-dm-server/ output/constraint_ summary.png", "best_ fit": { "m_chi_GeV": 986.0, "delta_keV": 350.0 } }
159	board post_finding	{"claim": "Tool compute_ relic_density (perturbative Kolb-Turner freeze-out, higgsino default sigma_v = 2.5e-26 cm^3/s x (1100 GeV/ m_chi)^2, x_f about 24.5, g* = 86.25) gives Omega h^ 2 = 0.0731, 0.0898, 0.1083, 0.1285, 0....	Recorded finding f-020.
160	board post_finding	{"claim": "The serve-idm.sh relic tool is perturbative only: it has no Sommerfeld or coannihilation switch, no charged-splitting input and no delta_m input (Omega h^ 2 = 0.1202 at 1160 GeV for delta = 400, 500 and 600 keV...	Recorded finding f-021.

#	Tool	Arguments	Outcome
161	board post_finding	{"claim": "check_all_constraints at m_chi = 1160 GeV returns 'VIABLE' for delta = 400, 500 and 600 keV, with relic Omega h^2 = 0.1202 (0.1 sigma). It gives N_expected_LZ = 0.00 (2.84 t-yr, sigma_n = 7.43e-39 cm^2), which...	Recorded finding f-022.
162	board post_question	{"text": "Why does check_all_constraints give N_LZ = 0.00 already at delta = 400 keV for m_chi = 1160 GeV, and why does it use a p-wave, splitting-suppressed sigma_v_today of about 1e-41 cm^3/s instead of the s-wave higg...	Recorded question q-15.
163	board post_finding	{"claim": "plot_constraint_summary on the 900-1300 GeV x 300-650 keV scan returns best fit m_chi= 986 GeV, delta=350 keV (plot /Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png); thi...	Recorded finding f-023.
164	board post_dead_end	{"what": "Quantifying the dependence of the delta_m= 500 keV rate on v_esc or an LMC high-velocity tail using the serve-idm tools.", "why": "predict_recoil, scan_mass_splitting and check_all_constraints have no halo param...	Recorded dead end x-03.
165	local cite	{"arxiv_id": "2609.01590", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib", "note": "[\"including the high-velocity tail generated by the gravita...	QUOTE FAILED: "At the joint best-fit mass, the thermal higgsino line prediction reaches 0.815-0" not found in source

#	Tool	Arguments	Outcome
166	board post_finding	{"claim": "2609.01590 (Wu, Zhang, Zhu) finds LZ-preferred delta_m = 481-519 keV (1 sigma) and 475-529 keV (2 sigma) for m_ chi = 1000-1183 GeV with the Halo 13 MW+LMC velocity distribution, but only 312-422 keV (1 sigma) ...	Recorded finding f-024.
167	board post_finding	{"claim": "The 475 keV and 557 keV neutrino lines differ because of how the captured population in the Sun is treated, not because of the halo. 2609.01590 simulates a non-thermal population on extended solar orbits at m_...	Recorded finding f-025.
168	board post_finding	{"claim": "Gamma-ray line status at about 1.1 TeV (2609.01590): at the joint best-fit mass the thermal higgsino line prediction is 0.815-0.900 of the H.E.S.S. observed limits (546 h), so it is close to but not excluded. ...	Recorded finding f-026.
169	board post_finding	{"claim": "Abstract-level only: 2610.03692 (Freese and Theodosopoulos) finds the 1.1 TeV thermal higgsino only marginally compatible with the LZ event and IceCube, and only with the LMC tail, no elastic solar cooling and...	Recorded finding f-027.
170	board post_question	{"text": "How large is the 2609.09830 one-loop change to delta_m and to the inelastic cross section (the full text has not been read), and what are the 2610.03692 thermal-mass IceCube variants in numbers?"}	Recorded question q-16.

#	Tool	Arguments	Outcome
171	local cite	{"arxiv_id": "2609.01590", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib", "note": "[\"including the high-velocity tail generated by the gravita...	Cited: TeV Higgsino Dark Matter from LZ Nuclear Recoil to Fermi-LAT Gamma Rays (\cite{arxiv260901590})
172	board post_dead_end	{"what": "Verbatim cite() of some 2609.01590 sentences: the kinematic-endpoint sentence 'controlled primarily by the kinematic endpoint v_esc + v_E', the HESS 0.815-0.900 ratio, the 553 keV equilibrium bound, the profile...	Recorded dead end x-04.
173	hep-research-aid load_table_data	{"filename": "/Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz"}	Error: 'utf-8' codec can't decode byte 0xc4 in position 15: invalid continuation byte
174	engine describe_tools	{}	{ "server": "inelastic-dm-server", "description": "Bridges particle physics (WIMP mass, mass splitting, cross section) with cosmological observables (relic density, recoil spectra, indirect detect...
175	serve-idm.sh scan_mass_splitting	{"m_chi_min": 1000, "m_chi_max": 1200, "delta_min": 330, "delta_max": 480, "n_points": 16}	{ "scan_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz", "grid": "16 x 16 = 256 points", "ranges": { "m_chi_GeV": [1000.0, 1200.0], ...

#	Tool	Arguments	Outcome
176	serve-idm.sh predict_recoil	{"m_chi_GeV": 1000, "delta_keV": 360}	{ "parameters": { "m_chi_GeV": 1000.0, "delta_keV": 360.0, "sigma_n_cm2": 7.428662722453626e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 320.8, "v_min_at_p...
177	serve-idm.sh predict_recoil	{"m_chi_GeV": 1183, "delta_keV": 360}	{ "parameters": { "m_chi_GeV": 1183.0, "delta_keV": 360.0, "sigma_n_cm2": 7.430820569964924e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 326.3, "v_min_at_p...
178	serve-idm.sh predict_recoil	{"delta_keV": 340, "m_chi_GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "delta_keV": 340.0, "sigma_n_cm2": 7.428662722453626e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 303.0, "v_min_at_p...
179	serve-idm.sh predict_recoil	{"delta_keV": 340, "m_chi_GeV": 1183}	{ "parameters": { "m_chi_GeV": 1183.0, "delta_keV": 340.0, "sigma_n_cm2": 7.430820569964924e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 308.2, "v_min_at_p...
180	local download_arxiv_full_text	{"arxiv_id": "2205.04486", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2205.04486.txt

#	Tool	Arguments	Outcome
181	serve-idm.sh predict_recoil	{"delta_keV": 380, "m_chi_GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "delta_keV": 380.0, "sigma_n_cm2": 7.428662722453626e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 338.7, "v_min_at_p...
182	serve-idm.sh predict_recoil	{"delta_keV": 380, "m_chi_GeV": 1183}	{ "parameters": { "m_chi_GeV": 1183.0, "delta_keV": 380.0, "sigma_n_cm2": 7.430820569964924e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 344.5, "v_min_at_p...
183	local download_arxiv_full_text	{"arxiv_id": "2609.01590", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.01590.txt
184	local cite	{"arxiv_id": "1807.06209", "note": "[\"0.120 \\\\ pm 0.001\"]", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib"}	QUOTE FAILED: "0.120 \ pm 0.001" not found in source
185	local download_arxiv_full_text	{"arxiv_id": "2610.03692", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2610.03692.txt

#	Tool	Arguments	Outcome
186	local cite	{"arxiv_id": "1912.07795", "note": "[\"less than 10^ -48 cm^2 in the TeV mass regime\"]", "bib_path": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244/ references.bib"}	QUOTE FAILED: "less than 10^-48 cm^2 in the TeV mass regime" not found in source
187	local download_arxiv_ full_text	{"arxiv_id": "2609.09830", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244/ 2609.09830.txt
188	hep-research-aid list_files	{"directory": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437"}	Files in /Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261016_110437: - 1104.0228.pdf - 1104.0228.txt - 1411.6930.pdf - 1411.6930.txt - 1807.06209.pdf - 1...
189	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244/ 2205.04486.txt", "start": 0, "length": 6000}	[chars 0-6000 of 118982] (more available, next start=6000) Title: The last Complex WIMPs standing Authors: Salvatore Bottaro, Dario Buttazzo, Marco Costa, Roberto Franceschini, Paolo Panci, Diego Red...
190	local read_arxiv_chunk	{"filepath": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-higgsino-dd/ runs/261022_110244/ 2609.09830.txt", "start": 0, "length": 12000}	[chars 0-12000 of 71943] (more available, next start=12000) Title: Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter Authors: Arindam Chatt...

#	Tool	Arguments	Outcome
191	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2610.03692.txt", "start": 0, "length": 12000}	[chars 0-12000 of 83465] (more available, next start=12000) Title: Higgsino dark matter compatible with the LUX-ZEPLIN high-energy nuclear-recoil event and IceCube constraints Authors: Katherine Free...
192	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt", "length": 3500}	[chars 0-3500 of 442128] (more available, next start=3500) Title: Planck 2018 results. VI. Cosmological parameters Authors: Planck Collaboration, N. Aghanim, Y. Akrami, M. Ashdown, J. Aumont, C. Bac...
193	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt", "length": 3500}	[chars 0-3500 of 37078] (more available, next start=3500) Title: Direct detection rate of heavy Higgsino-like and Wino-like dark matter Authors: Qing Chen, Richard J. Hill Year: 2019 arXiv ID: 1912.0...
194	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2205.04486.txt", "start": 6000, "length": 5000}	[chars 6000-11000 of 118982] (more available, next start=11000) 22 - 40 60 312 - 1.6 ×104 41/2 4.8± 0.3 ≃MPl 0.001 0.21 - 3 × 104 5× 106 20 - 1.9×104 51 9.9± 0.7 3 × 106 0.003 0.21 - 3 25 10 3- 2× 10...
195	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2610.03692.txt", "start": 12000, "length": 14000}	[chars 12000-26000 of 83465] (more available, next start=26000) arger Higgsino masses than 1.1 TeV and took into account the issues described above. However, there are two major differences between...

#	Tool	Arguments	Outcome
196	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.09830.txt", "start": 20000, "length": 14000}	[chars 20000-34000 of 71943] (more available, next start=34000) ced corrections play a crucial role in shaping the spin-independent elastic scattering relevant for direct detection experiments [44, 4...
197	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.09830.txt", "length": 9000, "start": 34000}	[chars 34000-43000 of 71943] (more available, next start=43000) t with colors signifying chargino mass splitting, δm ?$\chi\pm$ 1 =m ?$\chi\pm$ 1 -m ?χ0 1 . toh 1 as the mass ofh 2,m h2 >6 TeV, as compared tom h1$\sim$1...
198	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2610.03692.txt", "length": 16000, "start": 28000}	[chars 28000-44000 of 83465] (more available, next start=44000) thermal 1.1 TeV Higgsino better agreement with the combination of LZ data and IceCube bounds. In the next Sections, we take into accoun...
199	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt", "length": 2500, "start": 7000}	[chars 7000-9500 of 442128] (more available, next start=9500) index ns = 0.965$\pm$ 0.004, and optical depth τ = 0.054$\pm$ 0.007 (in this abstract we quote 68 % con?dence regions on measured parameters and ...
200	local cite	{"arxiv_id": "2205.04486", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib", "note": "[\"1.08$\pm$ 0.02\", \"obtained including Sommerfeld enhancement...	Cited: The last Complex WIMPs standing (\cite{arxiv220504486})

#	Tool	Arguments	Outcome
201	local cite	{"arxiv_id": "1912.07795", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib", "note": "[\"The direct detection signal in the pure Higgsino limit re...	Cited: Direct detection rate of heavy Higgsino-like and Wino-like dark matter (\cite{arxiv191207795})
202	serve-idm.sh predict_recoil	{"delta_keV": 350, "m_chi_GeV": 1000}	{ "parameters": { "m_chi_GeV": 1000.0, "delta_keV": 350.0, "sigma_n_cm2": 7.428662722453626e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 311.9, "v_min_at_p...
203	serve-idm.sh predict_recoil	{"delta_keV": 350, "m_chi_GeV": 1200}	{ "parameters": { "m_chi_GeV": 1200.0, "delta_keV": 350.0, "sigma_n_cm2": 7.430987655656208e-39, "is_higgsino": true }, "kinematics": { "E_R_peak_keV": 317.7, "v_min_at_p...
204	serve-idm.sh plot_constraint_summary	{}	{ "plot_file": "/Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png", "best_fit": { "m_chi_GeV": 1000.0, "delta_keV": 350.0 } }
205	board post_finding	{"claim": "2205.04486 (Bottaro et al., Table I) gives the thermal mass of the Dirac-fermion Y= 1/2 doublet (2_{1/2}, higgsino-like) as M_DM = 1.08 +/- 0.02 TeV, including Sommerfeld enhancement and bound-state formation, ...	Recorded finding f-028.

#	Tool	Arguments	Outcome
206	board post_finding	{"claim": "Chen and Hill (1912.07795) is cited in references.bib (key arxiv191207795) with the verbatim abstract sentence 'The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIM...	Recorded finding f-029.
207	board post_dead_end	{"what": "Citing Planck 2018 (1807.06209) with the quote '0.120 \\pm 0.001'.", "why": "Quote not found in the extracted text (PDF math garbling of Omega_c h^ 2); the abstract chunk read (chars 7000-9500) did not contain t...	Recorded dead end x-05.
208	board post_question	{"text": "The LZ 2024 (2410.17036) and XENONnT (2303.14729) SI limits at 1.1 TeV and the XLZD neutrino-fog level remain unread in d-11 (budget exhausted); can a follow-up read them from the text or HEPData instead of ext...	Recorded question q-17.
209	board post_finding	{"claim": "2609.09830 (Chatterjee et al.) gives no on-shell delta_m shift in the 300-600 keV range. Its benchmarks are tuned so the one-loop on-shell (CCN[n] scheme) neutral splitting is delta = 36.37, 8.90, 16.39, 10.81...	Recorded finding f-030.
210	board post_finding	{"claim": "2610.03692 (Freese and Theodosopoulos) shows four IceCube treatments at 90% CL as lower limits on delta in Fig. 2: capture-annihilation equilibrium; SI inelastic cooling only; SI inelastic + SD elastic (sigma_...	Recorded finding f-031.

#	Tool	Arguments	Outcome
211	board post_dead_end	{"what": "Re-verifying the f-024/f-025/f-026 numbers (481-519, 312-422 keV, 0.815-0.900, 553/557 keV, Delta chi^2 = 2.71) with narrow read_arxiv_ chunk windows, and doing the s-wave <sigma v> vs H.E.S.S. comparison.", "wh...	Recorded dead end x-06.
212	board post_finding	{"claim": "Under the tool's truncated SHM (v_max = 776 km/s), the LZ-compatible band (0.3-5 events, 2.84 t-yr) for m_chi = 1000-1200 GeV is confined to delta of about 350 keV. predict_ recoil gives N = 10.67/17.30 at delt...	Recorded finding f-032.
213	board post_finding	{"claim": "Analytic kinematics (m_N = 122.0 GeV for Xe-131, E_R = 248 keV, sqrt(2 m_N E_R) = 0.8206 keV per km/s): delta_max = v_ max*sqrt(2 m_N E_R) - m_N E_ R/mu gives, at m_chi = 1000/ 1100/1183 GeV, 358.5/361.3/ 363.2 ke...	Recorded finding f-033.
214	board post_finding	{"claim": "Inverting the kinematics at m_chi = 1100 GeV gives the v_max (Earth frame) needed to reach a given delta with an event exactly at E_R = 248 keV: 893 km/s for 457 keV, 915 km/s for 475 keV, 968 km/s for 519 keV...	Recorded finding f-034.
215	board post_dead_end	{"what": "load_table_data on /Users/nramachandra/ github-integration/ inelastic-dm-server/output/ scan_idm.npz to list the individual LZ-compatible scan points", "why": "load_ table_data only parses text/ CSV. The binary npz ...	Recorded dead end x-07.

A.5 Files produced

- 1104.0228.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1104.0228.txt
- 1411.6930.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1411.6930.txt
- 2609.09830.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2609.09830.txt
- 2410.17036.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2410.17036.txt
- 2606.31013.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/2606.31013.txt
- hep-ph_0610249.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/hep-ph_0610249.txt
- 1912.07795.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1912.07795.txt
- 1807.06209.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt
- 2609.01590.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.01590.txt
- 2609.07807.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.07807.txt
- 2205.04486.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2205.04486.txt
- 2610.03692.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2610.03692.txt
- 2609.09830.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/2609.09830.txt

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
d-01	closed	Establish the pure-higgsino thermal relic density as a function of mass (about 0.9, 1.0, 1.1, 1.2 and 1.3 TeV), with and without Sommerfeld enhancement, and find the mass at which Omega h^2 equals the Planck 2018 value of 0.120 +/- 0.001.	Not finished: only one published pure-higgsino relic-density point was found, so the success criteria (published values at three or more masses, and a thermal mass from two independent papers) are not met. (1) No relic-density or inelastic-DM compute tool exists here, and list_input_files returned no data files (tc-261016_1-w1-d-01-19, finding f-006). Every value below comes from the literature.
d-02	closed	Place the pure-higgsino elastic spin-independent cross section at about 1.0-1.2 TeV against the LZ 2024 (2410.17036) and XENONnT (2303.14729) limits at the same mass, and against the xenon neutrino fog.	Answer: a thermal pure higgsino at about 1.1 TeV lies about 2 or more orders of magnitude below the LZ 2024 limit and below the xenon neutrino floor. Three of the planned inputs were not read: the XENONnT limit, the LZ limit read from its figure at 1 TeV, and a numeric neutrino-fog value. All are recorded as open question q-06. (1) Prediction. Hisano et al. 1104.0228 work at leading order in elec
d-03	closed	Assess the routes by which a higgsino becomes visible to direct detection: the mass splittings and the inelastic Z-exchange threshold, LZ inelastic constraints (2609.09830), mixing-induced Higgs couplings in the (mu, M1/M2, tan beta) plane (2606.31013), and the (Omega h^2/0.12) rescaling for sub-thermal higgsinos.	Two of the three success criteria are met; the third is only partly met. (1) The inelastic threshold, worked by hand (f-001): m_Xe = 131 x 0.9315 = 122.0 GeV. The reduced mass mu_A is 108.7 GeV at m = 1.0 TeV and 109.8 GeV at 1.1 TeV. With v_max = 544 + 232 = 776 km/s, (v_max/c)^2 = 6.70e-6, so delta_max = mu_A v_max^2/2 is about 364 keV at 1.0 TeV and 368 keV at 1.1 TeV. The minimum-velocity form

Id	Status	Goal	Outcome
d-04	blocked	Pin down the pure-higgsino thermal mass from at least two independent published calculations, including recent work with bound-state formation, and quote the Planck 2018 Omega_c h^2 verbatim. Confirm which mass the 1411.6930 pure-higgsino values (0.126/0.127) refer to.	One of the three success criteria is met. (1) Met: the Planck 2018 value is quoted verbatim from the abstract on page 1 of arXiv:1807.06209: 'A combined analysis gives dark matter density Ωch2 = 0.120± 0.001' (68% CL). It is at roughly characters 6500-7000 of /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261016_110437/1807.06209.txt (finding f-012, tc-261016_1-w2-
d-05	blocked	Replace the linear 40 GeV extrapolation with direct reads of the LZ 2024 and XENONnT SI limits at about 1 TeV, a numeric xenon neutrino-fog value at about 1 TeV, and one next-generation projection (XLZD/DARWIN). Correct the f-008 arithmetic and quote Chen and Hill verbatim.	Two things were established: the Chen and Hill quotes and the corrected f-008 arithmetic. The 1 TeV limits, the neutrino-fog value and the XLZD projection were not obtained. (1) Chen and Hill (arXiv:1912.07795), verbatim from the Summary: 'The upper limit for the pure doublet is less than 10^-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than the est
d-06	closed	Verify 2606.31013 and 2609.09830 with metadata, and trace the contested f-002 and f-003 numbers to quoted excerpts: Eq. 13 (356 MeV), the f_q formula, Eq. 38, the Table VII NLO rows, and the tree-level delta_m0 formula (Eq. 9 / Ref. [28]).	Both papers are real, and every contested f-002/f-003 number except the tree-level delta_m0 formula is now traced to a verbatim excerpt. That formula is still unsourced, so the M1 needed for delta < 364 keV was not rederived. No cite() calls were made: the quotes are recorded on the board as f-015 and f-016, not in references.bib. (1) IDs confirmed by metadata (f-014, closes q-02). 2606.31013 is
d-07	open	Identify Ref. [28] of arXiv:2609.09830 and quote its Eq. (9) for the neutral splitting	
d-08	open	Add a sourced statement on indirect-detection limits for TeV higgsinos	

Id	Status	Goal	Outcome
d-09	closed	Use the serve-idm.sh tools to compute the pure-higgsino thermal relic density Omega h^2 at m_chi = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV, find the mass where Omega h^2 = 0.120, and run the combined constraint check at the thermal point.	Done. compute_relic_density (serve-idm.sh) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285 and 0.1503 at m_chi = 900, 1000, 1100, 1200 and 1300 GeV. It gives 0.1181 at 1150 GeV and 0.1202 at 1160 GeV, so the thermal mass (Omega h^2 = 0.120, Planck 0.12 +/- 0.0012) is about 1159 GeV, bracketed between 1150 and 1160 GeV (f-020). This is about 1 percent below the 1172 GeV inferred from 1411.6930 and
d-10	closed	Compute the inelastic Z-mediated recoil spectrum, the LZ event expectation and the (m_chi, delta_m) constraint map for the thermal higgsino. Reproduce or test the 2609.01590 preferred region (delta_m = 481-519 keV, m_chi = 1000-1183 GeV) against the 475 keV (IceCube) and 557 keV (Super-K + IceCube) neutrino lines.	All runs were local; get_dispatch reported 'local execution'. The tool uses a standard halo with a hard cutoff at v_esc+v_E = 776 km/s, and none of its calls accept a v_esc or LMC-tail option. At m_chi = 1100 GeV (sigma_n = 7.43e-39 cm^2), predict_recoil gives these values for delta = 300/400/480/520/560 keV: peak E_R = 270/360/432/468/504 keV, v_min at the peak = 701/809/886/922/957 km/s, and v_m
d-11	blocked	Close the remaining literature gaps: the 2205.04486 doublet thermal mass, the LZ 2024 and XENONnT SI limits near 1 TeV, and verbatim citations for Planck 2018 and Chen and Hill.	Two of the five gaps are closed; the other three are still open. (1) The doublet thermal mass from 2205.04486 (Bottaro et al., Table I) is quoted verbatim. For the Dirac-fermion Y=1/2 doublet it is M_DM = 1.08 +/- 0.02 TeV, computed with Sommerfeld enhancement and bound-state formation, normalised to Omega_DM h^2 = 0.11933 +/- 0.00091. The paper also says the doublet's direct-detection cross secti

Id	Status	Goal	Outcome
d-12	closed	Extract verbatim, citable inputs from the 2026 LZ-event papers: the halo models, rates and preferred regions, the neutrino exclusion lines, the radiative corrections and the rescue scenarios.	I read the full texts of 2609.01590 and 2609.07807 and only the abstracts of 2610.03692, 2609.09830 and 2609.06890. The validated citations are in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-higgsino-dd/runs/261022_110244/references.bib (keys arxiv260901590 and arxiv260907807). Not every number has a validated quote: the cite tool rejected several 2609.01590 sentences (the v_esc
d-13	closed	Run an analytic and tool-based kinematic check of delta_max(m_chi, v_max) at E_R = 248 keV. Identify the scan points compatible with the LZ event, and reconcile the tool's 776 km/s cutoff with the upper delta edges of 2609.01590 (profiled Maxwellian 422/457 keV; Halo 13 519/529 keV).	Main result: under the tool's halo, which cuts off at 776 km/s, no point that reproduces the LZ event has delta of 475 keV or more; the compatible band sits at about 350 keV. Everything ran locally. Points in the 0.3-5 event band were identified by bracketing with predict_recoil, because the scan file could not be read (x-07). Event counts from predict_recoil (LZ, 2.84 t-yr): - delta 340 keV: 10.
d-14	blocked	Recover quoted evidence for the contested literature numbers (f-024, f-025, f-026, f-027), and replace the tool's unphysical p-wave indirect-detection pass with an s-wave higgsino comparison at 1.1 TeV.	Two of the four goals are settled from full texts. The other two, plus the indirect-detection comparison, were not reached. (1) 2609.09830 does not give an on-shell delta_m shift for the 475-557 keV window (f-030). Its benchmark points are tuned to small one-loop on-shell splittings: delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5, at |mu| = 0.61-1.40 TeV, all with Omega h^2 < 0.12. Co

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	high	Kinematic threshold for inelastic up-scattering on xenon (my arithmetic): m_A = 131 x 0.9315 GeV = 122.0 GeV; mu_A = 108.7 GeV (m=1.0 TeV) and 109.8 GeV (m=1.1 TeV); v_max = 544+232 = 776 km/s, beta^2 = (776/299792)^2 = 6.70e-6; delta_max = mu_A beta^2 /2 = 364 keV (1.0 TeV) and 368 keV (1.1 TeV). The minimum-velocity formula v_min = sqrt(2 delta/mu_A) is given in 2609.09830 Appendix A (Eq. A7).	tc-261016_1-w1-d-03-42
f-002	high, **contested**	2609.09830 does not quote an LZ-excluded delta_m0 range. Its five benchmarks (|mu| = 0.61-1.40 TeV; |M1|, M2 about 5.8-42 TeV; tan beta 10.9-26.5) have one-loop on-shell splittings delta = 8.9-36.4 keV. Each predicts at most about 1.03 combined NLO events in LZ, and probing them needs 9.4-65.5 tonne-years. All are consistent with the LZ bound. Inelastic events at NLO range from 0.00022 (BP1) to 0.18 (BP4). NLO vertex corrections can exceed 100 percent, and in BP2 the inelastic cross section is about twice the LO elastic one. Splittings of O(10) keV come from partial cancellation of the M_Z^2/M	tc-261016_1-w1-d-03-39, tc-261016_1-w1-d-03-42, tc-261016_1-w1-d-03-29
f-003	high, **contested**	2606.31013: the SI coupling comes from Higgs exchange, f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + tan^2 theta_W/(M1-|mu|)](1 + sgn(mu) sin 2beta), so it is suppressed at large gaugino mass and at small tan beta with mu<0. The scan covers 150<|mu|<1000 GeV, 1.6<tan beta<50, 0.5<M1<10 TeV with M2=2M1. Rates are rescaled by Omega_H h^2/0.12 (Eq. 38, computed with micrOMEGAs 6.3.0). Direct detection is sensitive mainly where higgsino-gaugino mixing produces sizable charged-neutral splittings, and reaches the thermal mass of about 1 TeV. The radiative charged-neutral splitting tends to about 356 MeV for M >	tc-261016_1-w1-d-03-35, tc-261016_1-w1-d-03-30, tc-261016_1-w1-d-03-33 Fig. 4, 5

Id	Confidence	Claim	Evidence (tool calls, figures)
f-004	high	Hisano et al. (1104.0228) find the leading-order electroweak-loop SI cross section of pure EW-multiplet DM (including the higgsino, n=2, Y=1/2, evaluated at M=1 TeV) to be O(1e-46 to 1e-48) cm^2, suppressed by an accidental cancellation: twist-2 contributions are positive while Higgs-exchange, gluon and scalar contributions are negative, and they are of similar size. The cross section is nearly independent of mass above about 1 TeV.	tc-261016_1-w1-d-02-28, tc-261016_1-w1-d-02-34, tc-261016_1-w1-d-02-37 Fig. 5
f-005	high	LZ 2024 (2410.17036, 4.2 tonne-years): the strongest SI exclusion is 2.2e-48 cm^2 (90% CL) and the best median sensitivity is 5.1e-48 cm^2, both at 40 GeV/c^2. The abstract quotes no value at 1 TeV.	tc-261016_1-w1-d-02-49
f-006	high	No relic-density or inelastic-DM compute tool and no input data tables are available: list_input_files returned an empty file list, and the catalog contains only statistics/histogram, test-job and literature tools. All Omega h^2 values for d-01 therefore come from the literature and were not computed.	tc-261016_1-w1-d-01-19
f-007	medium, **contested**	Beneke, Hellmann and Ruiz-Femenia (arXiv:1411.6930, Sec. 3) report Omega h^2 = 0.127 (perturbative) and 0.126 (Sommerfeld, with off-diagonal rates) for a pure-higgsino model compared against the higgsino-like pMSSM point 1627006 at m_chi = 1172.31 GeV. The net Sommerfeld correction for pure higgsinos is almost zero (below 1 percent) because enhancement in the neutral channels cancels suppression in the chi0 chi+- channels. For the pMSSM higgsino-like point (higgsino fraction 0.98, dm+ = 1.8 GeV, dm0 = 9.5 GeV) they find Omega h^2 = 0.108 (perturbative) and 0.100 (Sommerfeld), an 8 percent redu	tc-261016_1-w1-d-01-27, tc-261016_1-w1-d-01-31, tc-261016_1-w1-d-01-40 Fig. 6

Id	Confidence	Claim	Evidence (tool calls, figures)
f-008	medium, **contested**	Chen and Hill (1912.07795) include the O(mW/M) power corrections, perturbative QCD to NNNLO and ChEFT nuclear effects. They find that the pure-higgsino SI cross section has an upper limit below 1e-48 cm^2 in the TeV mass range, below the estimated xenon neutrino floor. This is about 2 orders of magnitude below the LZ 2024 minimum limit of 2.2e-48 cm^2 at 40 GeV. Assuming linear mass scaling of the LZ limit above about 100 GeV (my estimate, not read from the paper), the LZ limit at 1.1 TeV is about 2.2e-48 x (1100/40), roughly 6e-47 cm^2, so the pure higgsino lies at least about 1.8 orders of m	tc-261016_1-w1-d-02-59, tc-261016_1-w1-d-02-58, tc-261016_1-w1-d-02-49
f-009	low	Hisano et al. (hep-ph/0610249) expect the Sommerfeld effect on the higgsino-like relic abundance to be roughly 10 percent, compared with a 50 percent reduction for the wino. 1411.6930 later found the net pure-higgsino effect is below 1 percent (Omega h^2 = 0.127 perturbative, 0.126 Sommerfeld). Rescaling that single point (taken as m = 1172.31 GeV) with Omega h^2 proportional to m^2 gives the Planck value 0.120 at m of about 1.14 TeV (Sommerfeld: 1144 GeV; perturbative: 1139 GeV). The rescaled values are 0.074, 0.092, 0.111, 0.132 and 0.155 at m = 0.9, 1.0, 1.1, 1.2 and 1.3 TeV. These are m^2	tc-261016_1-w1-d-01-68, tc-261016_1-w1-d-01-31, tc-261016_1-w1-d-01-66
f-010	high, **contested**	Chen and Hill (1912.07795), verbatim from the Summary: "The upper limit for the pure doublet is less than 10-48 cm2 in the TeV mass regime, consistent with the leading order estimate [18, 19], and lower than the estimated Xenon neutrino ?oor [26]." Verbatim from Sec. 4: "In the pure Higgsino limit, the cross section upper limit remains at or below 10 -48 cm2 for masses above a few hundred GeV." Abstract: "The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range." Sec. 4 also says that away from the pure limit, with c_H/g2^2 of ab	tc-261016_1-w2-d-05-76, tc-261016_1-w2-d-05-81

Id	Confidence	Claim	Evidence (tool calls, figures)
f-011	high	Correction to f-008: an upper bound below 1e-48 cm^2 compared with the LZ minimum of 2.2e-48 cm^2 (at 40 GeV) is a factor of more than 2.2 (more than 0.34 decades), not 2 orders of magnitude. The comparison that actually matters is at the same mass. At 1.1 TeV the linearly rescaled LZ limit of about 6e-47 cm^2, divided by the bound of 1e-48 cm^2, gives a ratio of at least about 60 (at least about 1.8 decades). That ratio is still an estimate: the LZ and XENONnT curves at 1 TeV were not read.	tc-261016_1-w2-d-05-81, tc-261016_1-w1-d-02-49
f-012	high	Planck 2018 (arXiv:1807.06209, abstract, page 1) states verbatim: "A combined analysis gives dark matter density $\Omega ch2 = 0.120 \pm 0.001$" (68% confidence region), quoted for base-LambdaCDM from polarization, temperature and lensing combined.	tc-261016_1-w2-d-04-82
f-013	medium	Bottaro et al., "The last Complex WIMPs standing" (arXiv:2205.04486, 2022), is the independent recent calculation of thermal masses for complex electroweak n-plets, which includes the Y=1/2 doublet (the higgsino). Its companion paper arXiv:2107.09688 states that the method includes "Sommerfeld enhancement and bound states formation at leading order", and 2205.04486 says the doublet with hypercharge 1/2 is the notable exception that future direct detection cannot fully probe. The numeric doublet thermal mass was not read in this direction.	tc-261016_1-w2-d-04-87, tc-261016_1-w2-d-04-89
f-014	high	Metadata confirms both IDs: arXiv:2606.31013 is 'Complementary Probes of Light Higgsinos: Electroweak Precision Measurements and Dark Matter Direct Detection' by Hamaguchi, Nagata and Osaki (2026), and arXiv:2609.09830 is 'Radiative Corrections to the Direct Detection of Inelastic Scattering of Higgsino-like Neutralino Dark Matter' by Chatterjee, Das, Pasha, Pukhov and Puri (2026). Both abstracts match how the survey uses them. This resolves q-02.	tc-261016_1-w2-d-06-70, tc-261016_1-w2-d-06-71

Id	Confidence	Claim	Evidence (tool calls, figures)
f-015	high	The full text of 2606.31013 supports the f-003 numbers. Eq. 13 reads Delta m+|rad ~ (1/2) alpha2 MZ sin^2 thetaW (1 - 3MZ/(2 pi M_chi+)), with the statement 'For M_chi+ >> MZ, this mass splitting asymptotically approaches Delta m+|rad ~ 356 MeV'. Eq. 33 reads f_q ~ -(g^2/8M_h^2)[1/(M2-|mu|) + tan^2 thetaW/(M1-|mu|)](1 + sgn(mu) sin 2beta) and is valid for MZ << M1,2-|mu|; the exact form is Eq. 32. The paper says the coupling is suppressed at large gaugino mass and 'for small tan beta and negative mu, due to an accidental cancellation'. Eq. 38 is sigma_SI/SD,eff = (Omega_H h^2/0.12) sigma_SI/SD	tc-261016_1-w2-d-06-77, tc-261016_1-w2-d-06-84, tc-261016_1-w2-d-06-80 Fig. 4, 5
f-016	high	The full text of 2609.09830 supports the f-002 numbers. Table III gives one-loop on-shell delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5, with mu = -739.63, -606.64, -814.26, 1282.11 and 1400.0 GeV, M1 = -7830, -6292, -5838, 11354 and 12875 GeV, M2 = 17055, 42000, 30925, 12627 and 12050 GeV, and tan beta = 10.88-26.46. Table VII, NLO rows for BP1-BP5: elastic events 1.0280, 0.07729, 0.1176, 0.4127, 0.5074; inelastic events 0.00022, 0.07039, 0.0518, 0.1806, 0.1661; combined 1.02822, 0.14768, 0.1694, 0.5933, 0.6735; required exposure 9.41, 65.50, 57.10, 16.30, 14.36 t-y (LZ exposure	tc-261016_1-w2-d-06-85, tc-261016_1-w2-d-06-90 Fig. 6
f-017	medium	2609.09830 does not write out the tree-level delta_m0 formula. It says only: 'as can be seen, for example, from Eq. (9) of Ref. [28], the corrections proportional to M_Z^2/M1 and M_Z^2/M2 depend on the relative signs of M1 and M2. In the parameter regions relevant for our analysis, these contributions can have opposite signs and partially cancel.' 2606.31013 also says only that the splittings are 'suppressed by M_Z^2/M_1,2'. The explicit form delta_m0 ~ m_Z^2 s_W^2/M1 + m_Z^2 c_W^2/M2 therefore remains unsourced (q-05 stays open), and so does the estimate that same-sign M2=2M1 needs M1 of abou	tc-261016_1-w2-d-06-85, tc-261016_1-w2-d-06-77

Id	Confidence	Claim	Evidence (tool calls, figures)
f-018	high	At m_chi=1100 GeV (higgsino, sigma_n=7.43e-39 cm^2, hard halo cutoff v_esc+v_E=776 km/s), predict_recoil gives v_min(248 keV)=701/823/921/969/1018 km/s and E_R,peak=270/360/432/468/504 keV for delta=300/400/480/520/560 keV, with N_LZ(extended NR, 2.84 t-yr)=744.6 at 300 keV and 0.00 for delta>=400 keV; the 2609.01590 window (481-519 keV) gives zero events under the tool halo, consistent with f-001 delta_max of about 365 keV.	tc-261022_1-w1-d-10-26, tc-261022_1-w1-d-10-27, tc-261022_1-w1-d-10-28, tc-261022_1-w1-d-10-29, tc-261022_1-w1-d-10-30, ...
f-019	medium	Local scan_mass_splitting (dispatch reported 'local execution') over m_chi=900-1300 GeV, delta=300-650 keV (15x15) finds 180/225 points pass all constraints and only 12 predict 0.3-5 LZ events; check_all_constraints at (1100 GeV, 500 keV) gives Omega h^2=0.1083, N_LZ=0.00 (not consistent with 1 event), sigma_v=1.29e-41 cm^3/s vs Fermi 2.3e-25, so 'viable' only because it is invisible; annual modulation at 248 keV is 0 (no rate).	tc-261022_1-w1-d-10-41, tc-261022_1-w1-d-10-42, tc-261022_1-w1-d-10-47, tc-261022_1-w1-d-10-46, tc-261022_1-w1-d-10-43
f-020	high	Tool compute_relic_density (perturbative Kolb-Turner freeze-out, higgsino default sigma_v = 2.5e-26 cm^3/s x (1100 GeV/m_chi)^2, x_f about 24.5, g* = 86.25) gives Omega h^2 = 0.0731, 0.0898, 0.1083, 0.1285, 0.1503 at m_chi = 900, 1000, 1100, 1200, 1300 GeV, and 0.1181 (1150 GeV) and 0.1202 (1160 GeV), so the thermal mass Omega h^2 = 0.120 is at about 1159 GeV (bracketed 1150-1160 GeV).	tc-261022_1-w1-d-09-21, tc-261022_1-w1-d-09-22, tc-261022_1-w1-d-09-23, tc-261022_1-w1-d-09-24, tc-261022_1-w1-d-09-25, ...
f-021	high	The serve-idm.sh relic tool is perturbative only: it has no Sommerfeld or coannihilation switch, no charged-splitting input and no delta_m input (Omega h^2 = 0.1202 at 1160 GeV for delta = 400, 500 and 600 keV). Its 'consistent_planck' flag is unreliable: it reports True at 25 sigma tension (1000 and 1300 GeV) and False at 39 sigma (900 GeV).	tc-261022_1-w1-d-09-18, tc-261022_1-w1-d-09-22, tc-261022_1-w1-d-09-25, tc-261022_1-w1-d-09-21, tc-261022_1-w1-d-09-38, ...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-022	medium, **contested**	check_all_constraints at m_chi = 1160 GeV returns 'VIABLE' for delta = 400, 500 and 600 keV, with relic Omega h^2 = 0.1202 (0.1 sigma). It gives N_expected_LZ = 0.00 (2.84 t-yr, sigma_n = 7.43e-39 cm^2), which is inconsistent with the 1 observed event. Indirect sigma_v_today = 6.7e-42, 1.04e-41 and 1.50e-41 cm^3/s, against a Fermi limit of 2.48e-25 cm^3/s. CMB p_ann = 0.86, 1.35 and 1.94 e-45, against a limit of 3.2e-28.	tc-261022_1-w1-d-09-38, tc-261022_1-w1-d-09-39, tc-261022_1-w1-d-09-40
f-023	medium	plot_constraint_summary on the 900-1300 GeV x 300-650 keV scan returns best fit m_chi=986 GeV, delta=350 keV (plot /Users/nramachandra/github-integration/inelastic-dm-server/output/constraint_summary.png); this lies below both the 475 keV (IceCube, 2609.01590) and 557 keV (Super-K+IceCube, 2609.07807) solar-neutrino exclusion lines, so under the tool's truncated SHM (v_max=776 km/s) no point reproducing the LZ event survives the neutrino limits, and the 2609.01590 region (481-519 keV) requires a high-velocity tail (e.g. LMC) not modelled by the tool.	tc-261022_1-w1-d-10-55, tc-261022_1-w1-d-10-42, tc-261022_1-w1-d-10-29, tc-261022_1-w1-d-10-30
f-024	high, **contested**	2609.01590 (Wu, Zhang, Zhu) finds LZ-preferred delta_m = 481-519 keV (1 sigma) and 475-529 keV (2 sigma) for m_chi = 1000-1183 GeV with the Halo 13 MW+LMC velocity distribution, but only 312-422 keV (1 sigma) and 280-457 keV (2 sigma), best fit m_chi = 1.09 TeV and delta_m = 366 keV, for a profiled Maxwellian. The text says the upper reach in delta_m is set by the kinematic endpoint v_esc + v_E. The relic mass range is 1000-1183 GeV (1 sigma) and 943-1230 GeV (2 sigma). LZ inputs: E_R = 248 +/- 23 stat +/- 23 sys keV and 2.84 tonne-year exposure.	tc-261022_1-w1-d-12-32, tc-261022_1-w1-d-12-36

Id	Confidence	Claim	Evidence (tool calls, figures)
f-025	high, **contested**	The 475 keV and 557 keV neutrino lines differ because of how the captured population in the Sun is treated, not because of the halo. 2609.01590 simulates a non-thermal population on extended solar orbits at m_chi = 1.11 TeV with no elastic cooling (elastic blind spot) and 10-year IceCube data, using <sigma v>_eff of about 1.5e-26 cm^3/s with Sommerfeld; this gives delta_m >~ 475 keV excluded at 90% CL (466 keV with loose solar/nuclear inputs, about 454 keV with the Fe nuclear-response uncertainty). If elastic scattering allows thermalization and capture-annihilation equilibrium, the same paper	tc-261022_1-w1-d-12-44, tc-261022_1-w1-d-12-33, tc-261022_1-w1-d-12-45, tc-261022_1-w1-d-12-37, tc-261022_1-w1-d-12-49
f-026	high, **contested**	Gamma-ray line status at about 1.1 TeV (2609.01590): at the joint best-fit mass the thermal higgsino line prediction is 0.815-0.900 of the H.E.S.S. observed limits (546 h), so it is close to but not excluded. The matched CTAO Alpha 500 h projection gives mu_gamma,95 = 1.69, which is above the thermal target. The estimate for the Omega configuration, 1.69/2 = 0.85, is below mu_gamma = 1, so the thermal target is within CTAO exclusion reach. Dwarf, CMB and ATLAS disappearing-track limits do not exclude the region.	tc-261022_1-w1-d-12-44, tc-261022_1-w1-d-12-36
f-027	medium, **contested**	Abstract-level only: 2610.03692 (Freese and Theodosopoulos) finds the 1.1 TeV thermal higgsino only marginally compatible with the LZ event and IceCube, and only with the LMC tail, no elastic solar cooling and no LZ high-energy sideband. Its compatible regions extend from about 2 TeV to hundreds of TeV, with delta about 350-525 keV. 2609.09830 computes one-loop EW corrections, with on-shell renormalization reducing delta_m and Higgs-exchange elastic and inelastic rates; the abstract gives no number for the size of the change. 2609.06890 proposes light boosted DM whose energy is set by the dark	tc-261022_1-w1-d-12-50, tc-261022_1-w1-d-12-51, tc-261022_1-w1-d-12-52

Id	Confidence	Claim	Evidence (tool calls, figures)
f-028	high, **contested**	2205.04486 (Bottaro et al., Table I) gives the thermal mass of the Dirac-fermion Y=1/2 doublet (2_{1/2}, higgsino-like) as M_DM = 1.08 +/- 0.02 TeV, including Sommerfeld enhancement and bound-state formation, with Omega_DM h^2 = 0.11933 +/- 0.00091; the paper also states the 2_{1/2} DD cross section lies well below the neutrino floor. This independent value is about 7 percent below the serve-idm.sh perturbative 1159 GeV (f-020).	tc-261022_1-w2-d-11-86, tc-261022_1-w2-d-11-91, tc-261022_1-w2-d-11-97
f-029	high	Chen and Hill (1912.07795) is cited in references.bib (key arxiv191207795) with the verbatim abstract sentence 'The direct detection signal in the pure Higgsino limit remains below neutrino backgrounds for WIMPs in the TeV mass range.'; the earlier board phrasing 'less than 10^-48 cm^2 in the TeV mass regime' was not found verbatim in the extracted text.	tc-261022_1-w2-d-11-83, tc-261022_1-w2-d-11-90, tc-261022_1-w2-d-11-98
f-030	medium, **contested**	2609.09830 (Chatterjee et al.) gives no on-shell delta_m shift in the 300-600 keV range. Its benchmarks are tuned so the one-loop on-shell (CCN[n] scheme) neutral splitting is delta = 36.37, 8.90, 16.39, 10.81 and 10.59 keV for BP1-BP5 (|mu| = 0.61-1.40 TeV; all have Omega h^2 < 0.12, so the relic constraint is not imposed). The tree-to-one-loop shift can be read from the Table III masses: about 0.14 GeV tree splitting vs 0 at one loop for BP1 (739.55/-739.69 to 739.33/-739.33 GeV), i.e. O(10-100) MeV scale. This paper therefore does not bear on the 475-557 keV window.	tc-261022_1-w2-d-14-94, tc-261022_1-w2-d-14-93
f-031	medium, **contested**	2610.03692 (Freese and Theodosopoulos) shows four IceCube treatments at 90% CL as lower limits on delta in Fig. 2: capture-annihilation equilibrium; SI inelastic cooling only; SI inelastic + SD elastic (sigma_SI,el = 0); SI inelastic + SD + SI elastic (sigma_SI,el = 4e-50 cm^2). It uses the W+W- annihilation-rate limits, an SHM with v0 = 220, v_sun = 232 and v_esc = 544 km/s, rho = 0.3 GeV/cm^3 and sigma_Hn = 7.43e-39 cm^2. For the solar bound it says the neutrino limits 'remain unchanged by adding the effects of the LMC'. For LZ, the SHM+LMC eta(v_min) is set to 0 above 950 km/s. The numerica	tc-261022_1-w2-d-14-95, tc-261022_1-w2-d-14-92, tc-261022_1-w2-d-14-88

Id	Confidence	Claim	Evidence (tool calls, figures)
f-032	high	Under the tool's truncated SHM (v_max = 776 km/s), the LZ-compatible band (0.3-5 events, 2.84 t-yr) for m_chi = 1000-1200 GeV is confined to delta of about 350 keV. predict_recoil gives N = 10.67/17.30 at delta = 340 keV (m_chi = 1000/1183 GeV), 1.13/2.90 at 350 keV (1000/1200 GeV), 0.03/0.158 at 360 keV (1000/1183 GeV) and 0.00/0.00 at 380 keV. A 16x16 scan (m_chi 1000-1200 GeV, delta 330-480 keV in 10 keV steps) reports exactly 16 points in the 0.3-5 band, best fit (1000 GeV, 350 keV). No LZ-compatible point has delta >= 475 keV, or even >= 360 keV.	tc-261022_1-w2-d-13-74, tc-261022_1-w2-d-13-75, tc-261022_1-w2-d-13-76, tc-261022_1-w2-d-13-77, tc-261022_1-w2-d-13-78, ...
f-033	high	Analytic kinematics (m_N = 122.0 GeV for Xe-131, E_R = 248 keV, sqrt(2 m_N E_R) = 0.8206 keV per km/s): delta_max = v_max*sqrt(2 m_N E_R) - m_N E_R/mu gives, at m_chi = 1000/1100/1183 GeV, 358.5/361.3/363.2 keV for v_max = 776 km/s, 378/381/383 keV for 800 km/s, 419/422/424 keV for 850 km/s and 460/463/465 keV for 900 km/s. This reproduces the tool's v_min(248 keV) to within 1 km/s (766 vs 765.7 km/s at 1000 GeV and 350 keV; 921/969/1018 km/s at 1100 GeV and 480/520/560 keV). The tool's 'about 365 keV' edge is the E_R-optimized endpoint delta_max = mu v_max^2/2 = 364/368/370 keV at 776 km/s, r	tc-261022_1-w2-d-13-75, tc-261022_1-w2-d-13-76, tc-261022_1-w2-d-13-99, tc-261022_1-w2-d-13-100
f-034	medium	Inverting the kinematics at m_chi = 1100 GeV gives the v_max (Earth frame) needed to reach a given delta with an event exactly at E_R = 248 keV: 893 km/s for 457 keV, 915 km/s for 475 keV, 968 km/s for 519 keV, 980 km/s for 529 keV and 1015 km/s for 557 keV (about +3 km/s at 1000 GeV, about -2 km/s at 1183 GeV). If E_R is free (at the optimal E_R = mu*delta/m_N of about 380-500 keV), the minimum is v_max = sqrt(2 delta/mu): 830 km/s for 422, 864 for 457, 881 for 475, 921 for 519, 930 for 529 and 954 km/s for 557 keV. All of these exceed the tool's SHM cutoff of 776 km/s by 55-240 km/s. So the	tc-261022_1-w2-d-13-75, tc-261022_1-w2-d-13-81, tc-261022_1-w2-d-13-99

B.3 Dead ends

- `x-01` (`d-05`): Reading the LZ 2024 (2410.17036) and XENONnT (2303.14729) SI limits at 1 TeV with get_paper_figures — The HTML extraction returned none of the limit plots. For LZ it returned only Fig. 1 (fiducial-volume data) and Fig. S4 (salt distribution), registered as Figures 10 and 11. For XENONnT it returned only calibration and spatial-distribution plots, registered as Figures 12-14. The LZ text chunk read (characters 20000-28000) contained no 1 TeV value. A numeric 1 TeV limit needs the HEPData tables or

- `x-02` (`d-04`): Searching abstracts for a recent higgsino thermal mass with the query 'last complex WIMPs standing electroweak multiplet thermal mass bound states'. — The search returned only unrelated LEP electroweak-precision notes. The relevant paper (2205.04486) was found from its known arXiv ID through get_arxiv_metadata, so direct ID lookup is the better route.

- `x-03` (`d-10`): Quantifying the dependence of the delta_m=500 keV rate on v_esc or an LMC high-velocity tail using the serve-idm tools. — predict_recoil, scan_mass_splitting and check_all_constraints have no halo parameters; v_esc+v_E is hardcoded at 776 km/s, so every point with delta >= about 370 keV at 1.1 TeV returns exactly 0 events. A custom rate code with a variable halo is needed.

- `x-04` (`d-12`): Verbatim cite() of some 2609.01590 sentences: the kinematic-endpoint sentence 'controlled primarily by the kinematic endpoint v_esc + v_E', the HESS 0.815-0.900 ratio, the 553 keV equilibrium bound, the profiled Maxwellian best fit and the relic mass ranges. — The PDF text extraction mangles spacing around math symbols, and the en-dash ranges failed to match, so these quotes did not validate. The numbers were read in tc-...-32, -36 and -44. Only short phrases without math (LMC tail, 475 keV, 466 keV, 'close to, but not excluded by, the present H.E.S.S. limits') were cited successfully.

- `x-05` (`d-11`): Citing Planck 2018 (1807.06209) with the quote '0.120 \pm 0.001'. — Quote not found in the extracted text (PDF math garbling of Omega_c h^2); the abstract chunk read (chars 7000-9500) did not contain the Omega_c h^2 sentence. A shorter fragment from the earlier abstract part (before char 7000) must be located and retried. Planck is not yet in references.bib.

- `x-06` (`d-14`): Re-verifying the f-024/f-025/f-026 numbers (481-519, 312-422 keV, 0.815-0.900, 553/557 keV, Delta chi^2 = 2.71) with narrow read_arxiv_chunk windows, and doing the s-wave <sigma v> vs H.E.S.S. comparison. — The tool budget ran out after re-downloading 2609.01590. No chunk of 2609.01590 or 2609.07807 was read in this direction, so those findings remain contested, and no prediction-to-limit ratio for WW/ZZ continuum was obtained. Whether the 1.69/2 CTAO Omega estimate comes from the paper or from the agent is also still unverified.

- `x-07` (`d-13`): load_table_data on /Users/nramachandra/github-integration/inelastic-dm-server/output/scan_idm.npz to list the individual LZ-compatible scan points — load_table_data only parses text/CSV. The binary npz fails with a utf-8 decode error, and no tool returns the per-point scan grid, so band membership has to be inferred from predict_recoil bracketing.

B.4 Critic verdicts

- `c-01` (wave 0, plan): **revise**; high: Step 1 is not enough: the gap has to show up in the final answer. No relic-density or inelastic-DM compute tool exists, list_input_files is

empty, and run_test_job only draws random samples. The task asked us to compute relic density with tools, and we cannot. The report must say plainly that every ; high: 1411.6930 is a pMSSM scan. It may not give a pure-higgsino Omega h^2(m) curve at 0.9-1.3 TeV with and without Sommerfeld enhancement. Before relying on it, check that such a figure or table exists. If it does not, use the one-parameter thermal mass plus the m^2 scaling, and label every resulting row; medium: The budget is tight. Fifteen calls do not cover three or four full-text downloads plus chunk reads, figures and cite. Drop the full read of 1008.2905, which is a method paper and not needed for numbers. Get the Planck Omega_c h^2 = 0.120 +/- 0.001 quote with one targeted chunk read of 1807.06209. Li; medium: Run get_arxiv_metadata on 2606.31013 and 2609.09830 before any download, to confirm they exist and say what the brief claims. The brief cites 2606.31013 for the '~1 TeV' figure but has not read it. If either ID does not resolve, report that and do not attribute numbers to it.; high: A key literature cross-check is missing. Hill and Solon (for example 1111.0016 and 1409.8290) give the heavy-higgsino SI cross section with a large cancellation-driven uncertainty, possibly near or below the xenon neutrino fog. Because 1104.0228 and 1007.2601 predate refined nucleon matrix elements,; medium: Values read off log-scale limit figures with get_paper_figures need a stated reading uncertainty, for example a factor of about 1.5. Check whether the LZ and XENONnT figures actually extend to 1 TeV. If they do, read the value there and do not rescale. If rescaling is needed, use limit proportional

- `c-02` (wave 1, wave): **continue**; contested `f-003`, `f-007`, `f-008`, `f-002`; high: The core quantity is still unmet. There is no published Omega h^2(m) curve at multiple masses and no second, independent determination of the thermal mass. The m^2 rescaling in f-009 depends on an inferred mass of 1172.31 GeV. If that inference is wrong, every rescaled value and the 1.14 TeV thermal; medium: The Planck 2018 value of 0.120 +/- 0.001 is used throughout but has never been quoted from 1807.06209.txt, although the file is already downloaded. 1411.6930 normalised to Planck 2013 (0.1187), so the f-009 rescaling should state which target it uses.; low: The arithmetic checks out: 0.126 x (m/1172.31)^2 gives 0.074, 0.092, 0.111, 0.132 and 0.155, and the thermal masses of 1144 and 1139 GeV are correct. However, the hep-ph/0610249 '10 percent' statement is not visible in the tc-...-68 excerpt.; low: The excerpt confirms only the 4.2 t-y exposure. The 2.2e-48 / 5.1e-48 cm^2 values at 40 GeV match the published LZ abstract but are not visible in the quoted chunk.; low: The excerpts support the structure of the cancellation (the Fig. 6 contribution panels), but not the numeric range O(1e-46 to 1e-48) cm^2. This is a 2011 result computed against a scan over the Higgs mass, so it should be quoted at mh = 125 GeV.; high: The headline answer to 'where the higgsino sits relative to limits' still rests on a linear extrapolation from 40 GeV. None of the following has been read: the LZ or XENONnT limit at 1 TeV, a numeric neutrino-fog value, or projected sensitivities (LZ full exposure, XLZD). The final answer needs thes

- `c-03` (wave 2, wave): **continue**; contested `f-010`; medium: The arithmetic is correct: 2.2e-48 / 1e-48 = 2.2, which is 0.34 decades, and 2.2e-48 x 1100/40 = 6.05e-47, giving a ratio of at least about 60 (about 1.78 decades). The key number is still a linear extrapolation from the 40 GeV minimum, not a read value. The 2.2e-48 input comes from f-005 and is not; low: The excerpt from tc-...-82 is truncated before the quoted sentence. The retrieved range (chars 5500-10000) does cover the stated location (about 6500-7000), and the value is standard, so it is accepted. It is still not in references.bib; cite() should be called.; medium: None of the Table III, VI or VII numbers appear in the displayed excerpts. Only surrounding prose is shown, although the retrieved ranges (chars 27000-50000) plausibly contain the tables. This was the original reason f-002 was contested, so the trace is only partly verbatim. f-016 is accepted on the; medium: The tree-level estimate that same-sign

M2 = 2 M1 needs M1 of about 1.4e7 GeV is misleading for the survey's conclusion. A same-sign tree-level splitting of order m_Z^2 (s_W^2/M1 + c_W^2/M2) at about 12 TeV would be roughly 0.5-0.7 MeV. BP4 and BP5 instead show about 10.6-10.8 keV after on-shell one-; medium: No relic-density or inelastic-DM compute tool exists (f-006), so the task's 'compute the relic density at several masses' cannot be done as stated. The final answer must state this plainly. The m^2 rescaling in f-009 should be presented as an interpolation that depends on the unverified 1172.31 GeV ; low: No cite() calls were made for the Chen and Hill quotes, Planck, or the 2606.31013 and 2609.09830 traces in this wave, so references.bib does not reflect the evidence on the board.

- `c-04` (wave 2, report): **ok**

- `c-05` (wave 0, plan): **revise**; high: The steps need about 17 calls: 1 describe, 5 masses, 3 toggles, up to 3 bracketing, 4 sensitivity, 1 constraint check. The budget is 15. Drop the 400 and 600 keV delta_m variations, since a keV-scale neutral splitting has a negligible effect on freeze-out. Keep one delta_m variation as a sanity chec; high: If describe_tools shows that compute_relic_density is perturbative only, or omits coannihilation, the crossing will land near 1.0 TeV or lower. It must then not be reported as the higgsino thermal mass. State the result as tool output under those stated assumptions, and compare it with 1411.6930 and; medium: The 1172 GeV value for 1411.6930 is inferred, not stated in the paper. Label it as inferred everywhere. For check_all_constraints, record whether its LZ test is elastic SI, inelastic, or both, and which exposure it assumes. Without this, the LZ verdict cannot be interpreted.; high: Do not call set_dispatch to polaris or perlmutter. No IRI or Globus authentication has been established, and get_dispatch reports local execution. Size the grid so it runs locally, for example 5 masses by 8 delta_m values, and report the site as returned.; high: Event counts are unverifiable without LZ signal efficiency and the analysis region of interest near 248 keV, since the standard WIMP search region ends well below that. State the acceptance the tool uses. If the tool has none, report counts as 100 percent acceptance upper bounds and label them that ; medium: f-001 checks out: mu of about 110 GeV and v of 776 km/s give delta_max of about 368 keV. A standard halo therefore gives zero events at 480-520 keV, and the result rests entirely on halo assumptions. Make the v_esc or high-velocity-tail variation mandatory, not conditional. If the tool has no such k

- `c-06` (wave 1, wave): **continue**; contested `f-022`, `f-024`, `f-025`, `f-026`, `f-027`; high: The central task question has not been answered quantitatively: where does the elastic SI cross section of the thermal higgsino sit relative to the LZ 4.2 t-yr and XENONnT 90% CL curves at 1.0-1.2 TeV? No finding gives either the higgsino SI value (loop and Higgs-mediated, around 1e-48 to 1e-49 cm^2; medium: The thermal mass of about 1159 GeV is set by the tool's hard-coded normalization (sigma_v = 2.5e-26 x (1100/m)^2). It is a calibration, not a computation. The report must not present it as an independent check of 1411.6930. The Sommerfeld/coannihilation step and the charged-splitting sensitivity in ; medium: The tool's indirect-detection and CMB passes rest on a p-wave, splitting-suppressed annihilation today. This is unphysical for a higgsino, which annihilates s-wave to WW/ZZ at about 1-1.5e-26 cm^3/s. The 'VIABLE' label for indirect and CMB constraints should be replaced by a comparison of the s-wave; medium: The claim that no LZ-reproducing point survives the neutrino limits is inferred from the 1.1 TeV kinematics and the single best fit. The 12 'best_fit_region' points were never listed. The npz file exists and should be loaded to confirm their (m_chi, delta) values. A second point needs checking: the ; low: Several headline literature numbers rest on text that was read but has no validated quote (x-04). Re-fetching narrower character windows around the relevant passages would let the excerpts carry the values even if the cite tool continues to reject garbled math.

- `c-07` (wave 2, wave): **continue**; contested `f-028`, `f-030`, `f-031`; high: The core comparison asked for in the task is still missing after two waves. No LZ 4.2 t-yr or XENONnT SI 90% CL limit near 1.0-1.2 TeV has been read. The higgsino elastic sigma_SI has not been placed against those limits or the neutrino fog, and Planck 2018 is uncited. The final answer cannot be sta; medium: The 'compatible band at about 350 keV' is an artefact of the tool's hard 776 km/s cutoff. N falls from 10.7 to 1.1 to 0.03 over 20 keV. This should be stated as a property of the truncated SHM, not as a physical preferred region. The scan also overwrote the d-10 scan_idm.npz and constraint_summary.p; low: The kinematics assume a single isotope with m_N = 122 GeV (Xe-131). Natural xenon spans A = 124-136. For example, Xe-134 raises delta_max at 776 km/s and 1.1 TeV by about 7 keV. The 248 +/- 23 stat +/- 23 sys keV energy uncertainty also matters. The v_max thresholds therefore carry roughly +/- 10-20; medium: The tool's p-wave indirect-detection and CMB passes are still the only indirect numbers on the board, and they are unphysical for an s-wave higgsino. f-026 (0.815-0.900 of H.E.S.S., CTAO reach) also remains contested.

- `c-08` (wave 2, report): **ok**; high: unsupported: LZ reported a nuclear-recoil candidate at E_R = 248 ± 23 stat ± 23 sys keV, with 3.4σ local and 2.6σ global significance, in a 2.84 tonne-year exposure (arXiv:2609.06890, arXiv:2609.02608). — The board sources the event properties to arXiv:2609.02608 and the ±23 keV uncertainties to arX; high: unsupported: Kolb–Turner freeze-out (x_f ≈ 24.4–24.8, g* = 86.25) — The board records only 'x_f about 24.5'. The 24.4–24.8 range is more specific than any recorded tool output. This is minor.; high: unsupported: Several literature numbers (preferred regions, 553 keV, the H.E.S.S. ratio, the CTAO Omega estimate, the doublet mass, ...) were not independently verified. The critic contested them, mostly because of failed verbatim quotes or abstract-only reading. — d-11 states that the 2205.04486 do; high: unsupported: 7. Add isotope averaging over natural xenon and the ±23 keV energy uncert — The report is truncated mid-sentence, so the final open direction is incomplete. Its content matches critique c-07, but the sentence needs to be finished.

C Figures not shown in the main text

Produced during the session; captions carry their provenance.

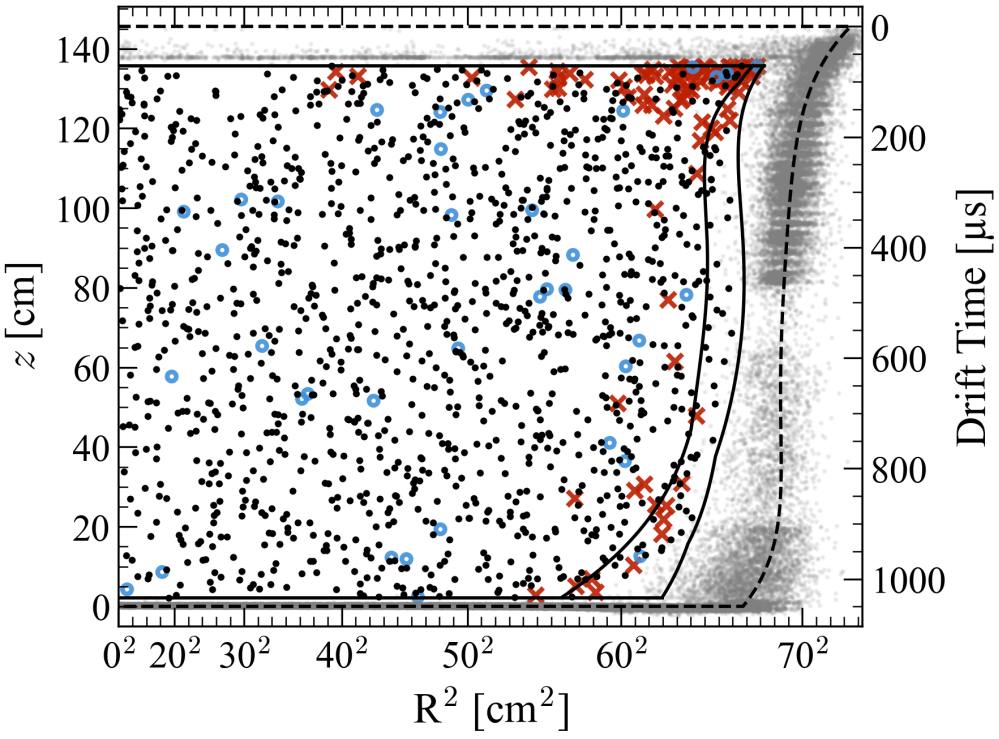


Figure 1: Figure 1: Data from WS2024, in black with all analysis cuts applied, and in gray outside the FV without veto coincidence cuts. Red crosses and blue circles represent events that are vetoed by the prompt and delayed veto coincidence cuts, respectively. The dashed line shows the active volume, averaged over azimuth, and the two solid lines depict the FV, at the azimuths of its smallest and largest radial extent. The drift field affects the position reconstruction such that the observed radius of the TPC varies as a function of both drift time and azimuthal angle. [arXiv:2410.17036] Reproduced from arXiv:2410.17036 (extracted with `get_paper_figures`).

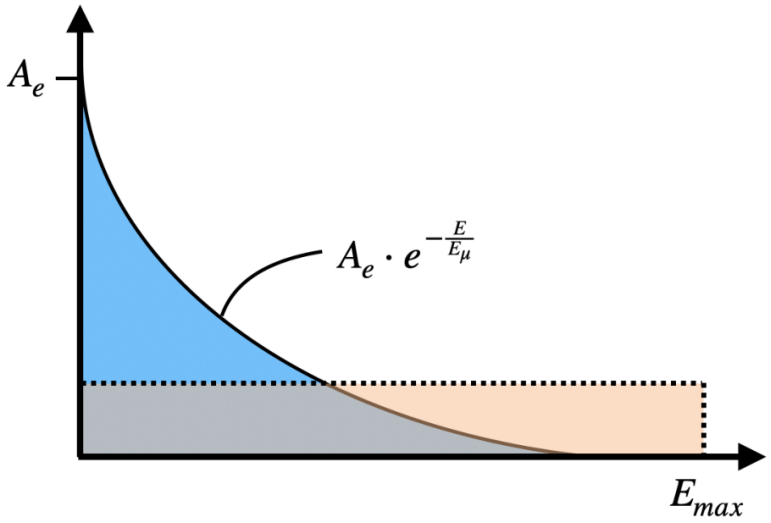


Figure 2: Figure S4: Distribution from which the salt events are drawn. The parameters are described in the text. [arXiv:2410.17036] Reproduced from arXiv:2410.17036 (extracted with `get_paper_figures`).

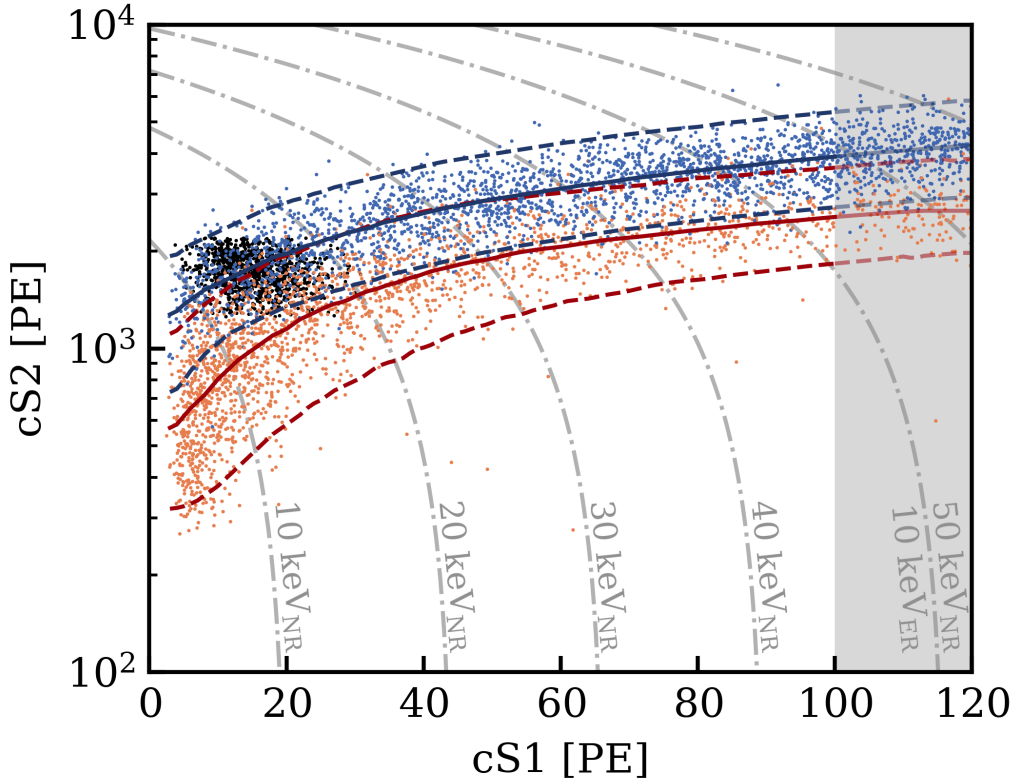


Figure 3: Figure 1: NR and ER calibration data from {}^{\makebox[11.95839pt][r]{$\scriptstyle 241$}}_{\makebox[11.95839pt][r]{$\scriptstyle$}}\mathrm{AmBe} (orange), {}^{\makebox[11.95839pt][r]{$\scriptstyle 220$}}_{\makebox[11.95839pt][r]{$\scriptstyle$}}\mathrm{Rn} (blue) and {}^{\makebox[7.97226pt][r]{$\scriptstyle 37$}}_{\makebox[7.97226pt][r]{$\scriptstyle$}}\mathrm{Ar} (black). The median and the \pm 2\text{\,}\sigma contours of the NR and ER model are shown in blue and red respectively. The gray dash-dotted contour lines show the reconstructed NR energy (\mathrm{$\mathrm{k}\mathrm{e}\mathrm{V}${} [arXiv:2303.14729] Reproduced from arXiv:2303.14729 (extracted with `get_paper_figures`).

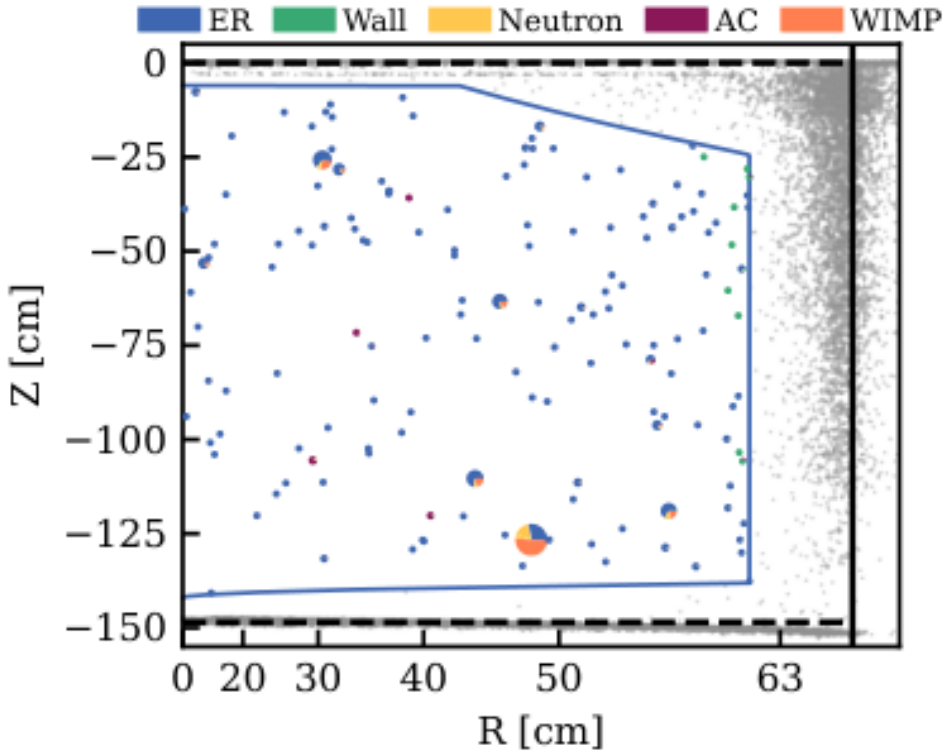


Figure 4: Figure A.2: Spatial distribution of the search data in the 4.18\text{\,}\mathrm{t} fiducial volume (blue line). Each event is represented with a pie-chart, showing the fraction of the best-fit PDF including a 200\text{\,}\mathrm{GeV}\text{/}{\mathrm{\text{c}}}^{2} WIMP evaluated at the position of the event, color-coded as in Figure 3. Events reconstructed outside of the fiducial volume are colored in gray. Black dashed lines depict the boundaries of the sensitive volume given by the cathode and gate positions. The TPC radius is indicated by a vertical black line. [arXiv:2303.14729] Reproduced from arXiv:2303.14729 (extracted with `get_paper_figures`).

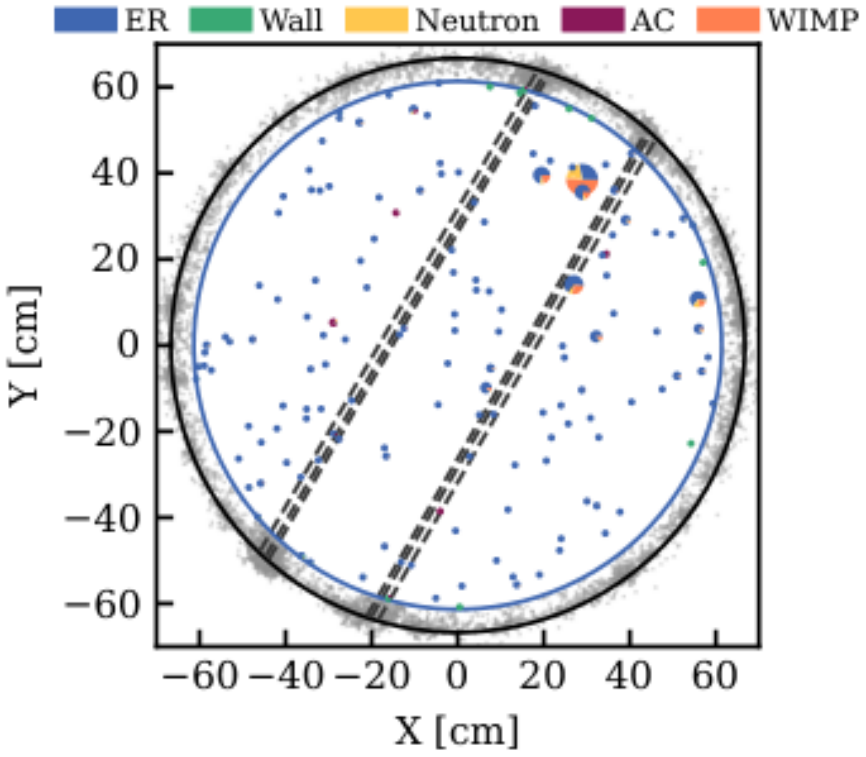


Figure 5: Figure A.3: Distribution of the search data in the horizontal plane. Each event is represented with a pie-chart, showing the fraction of the best-fit PDF including a 200\text{\,}\mathrm{GeV}\text{/}{\mathrm{\text{c}}}^{2} WIMP evaluated at the position of the event, color-coded as in Figure 3. Events with a reconstructed radius outside of the fiducial volume are shown in gray. The inner (outer) circle indicates the FV selection (TPC radius), and dashed lines show the transverse wire positions. [arXiv:2303.14729] Reproduced from arXiv:2303.14729 (extracted with `get_paper_figures`).